

Convergence & Divergence

Maclaurin's & Taylor

Leibnitz Theorem

L-Hopital, sequence

Limits & continuity.

Section A & B practice

TOPIC: LIMITS & CONTINUITY

LIMITS & CONTINUITY:-

$$S = \{6, 7, 8, 9, 10, 11, 12\}$$

- Lower bound of a set are $6, 5, 4, 3, 2, \dots$
- Upper bound of a set are $12, 13, 14, 15, 16, \dots$
- Greatest lower bound of a set is 6 or $m=6$
- Least upper bound of a set is 12 or $M=12$

BOUND SET:-

A set 'A' is called bounded set if it has both upper and lower bounds

e.g. $\{1, 2, \dots, 5\}$

$$\{10, 20, 30, \dots, 100\}$$

UNBOUND SET:-

A set 'A' is called Unbounded set if it has neither upper bound nor lower bound.

e.g. set of integers.

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

BOUNDED ABOVE SETS:-

A set is called bounded above set if it has only upper bounds.

e.g. $\{-1, -2, -3, \dots\}$
 $\{10, 9, 8, 7, \dots\}$

The order of bounded above set is always in decreasing order.

BOUNDED BELOW SETS:-

A set is called bounded below set if it has only lower bounds. set will be in increasing order.

e.g. set of natural numbers, whole numbers etc

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

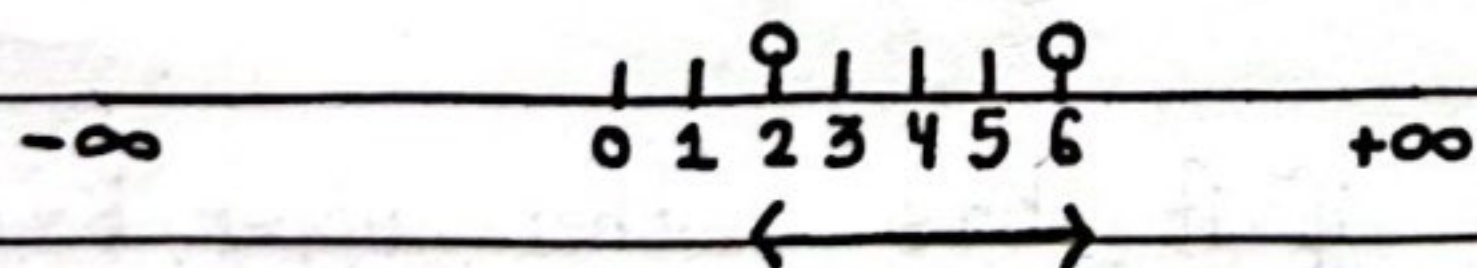
$$\mathbb{W} = \{0, 1, 2, 3, \dots\}$$

OPEN INTERVAL :-

$$\text{Bounded Set} = (2, 6)$$

$$(a, b) = \{x/a < x < b\}$$

$$(2, 6) = \{x/2 < x < 6\}$$



$$A = (2, 6)$$

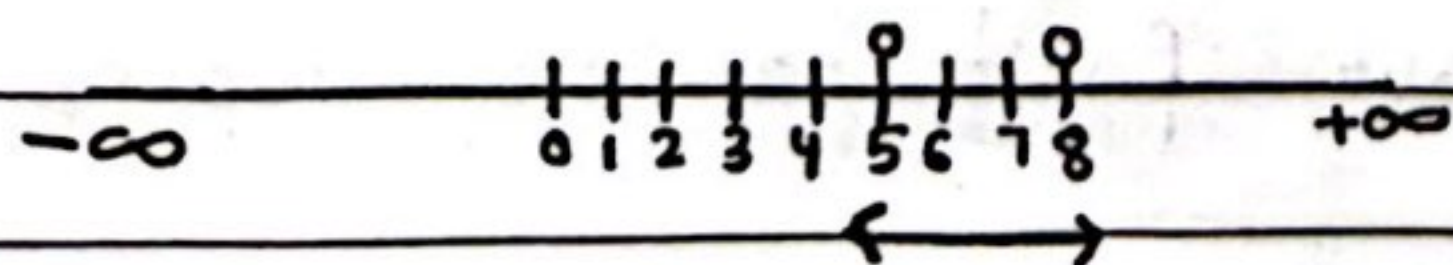
$$m = 2 \notin A$$

$$M = 6 \notin A$$

CLOSED INTERVAL:

$$[a, b] = \{x/a \leq x \leq b\}$$

$$[5, 8] = \{x/5 \leq x \leq 8\}$$



$$A = [5, 8]$$

$$m = 5 \in A$$

$$M = 8 \in A$$

SEMI OPEN / SEMI CLOSED INTERVALS:-

$$(a, b] = \{x/a < x \leq b\}$$

$$[a, b) = \{x/a \leq x < b\}$$

$$A = (3, 9]$$

$$B = [2, 5)$$

FUNCTIONS:-

If a variable y depends on a variable x in such a way that each value of x determine exactly one value of y , then we say that y is a function of x .
Four common methods for representing functions are

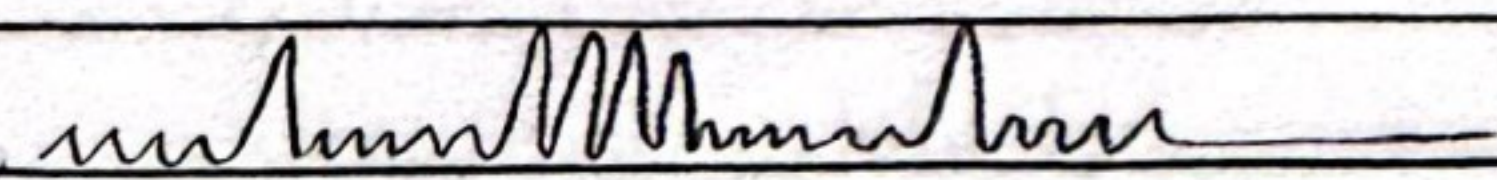
- 1 Numericals by tables
- 2 Geometrically by graphs
- 3 Algebraically by formula
- 4 Verbally

NUMERICAL:-

Autotrace, Qualifying speeds

year, t	speed, S
1994	228.011
1995	231.604
1996	233.100

GEOMETRICALLY:-


graphically Earthquake

ALGEBRAICALLY:-

Functions arises from formulas

$c = 2\pi r$, circumference of circle as a function of r .

VERBALLY:-

Sometimes functions are describe in words.
e.g. of Universal Gravitation

$$F = \frac{Gm_1m_2}{r^2}$$



r^2

Gravitational force of attraction
(F as a function of r)

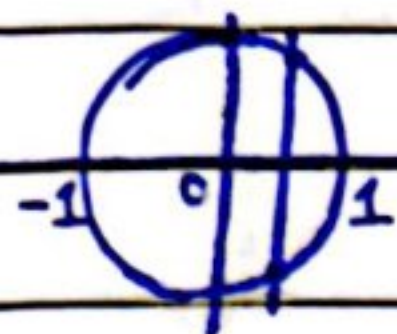
$$y = f(x)$$

dependent Independent

$$f(x) = x$$

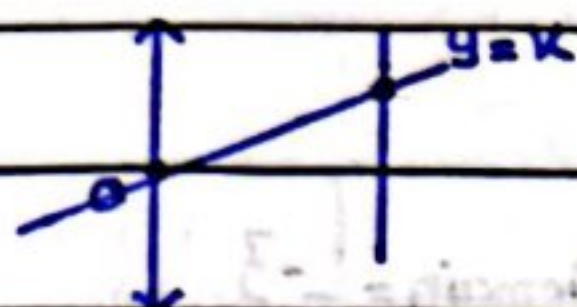
function Input output
name

VERTICAL LINE TEST:-



$$x^2 + y^2 = 1$$

It's not a function



It's a function.

THE ABSOLUTE VALUE FUNCTION:-

$$|x| = \begin{cases} x & , x \geq 0 \\ -x & , x < 0 \end{cases}$$

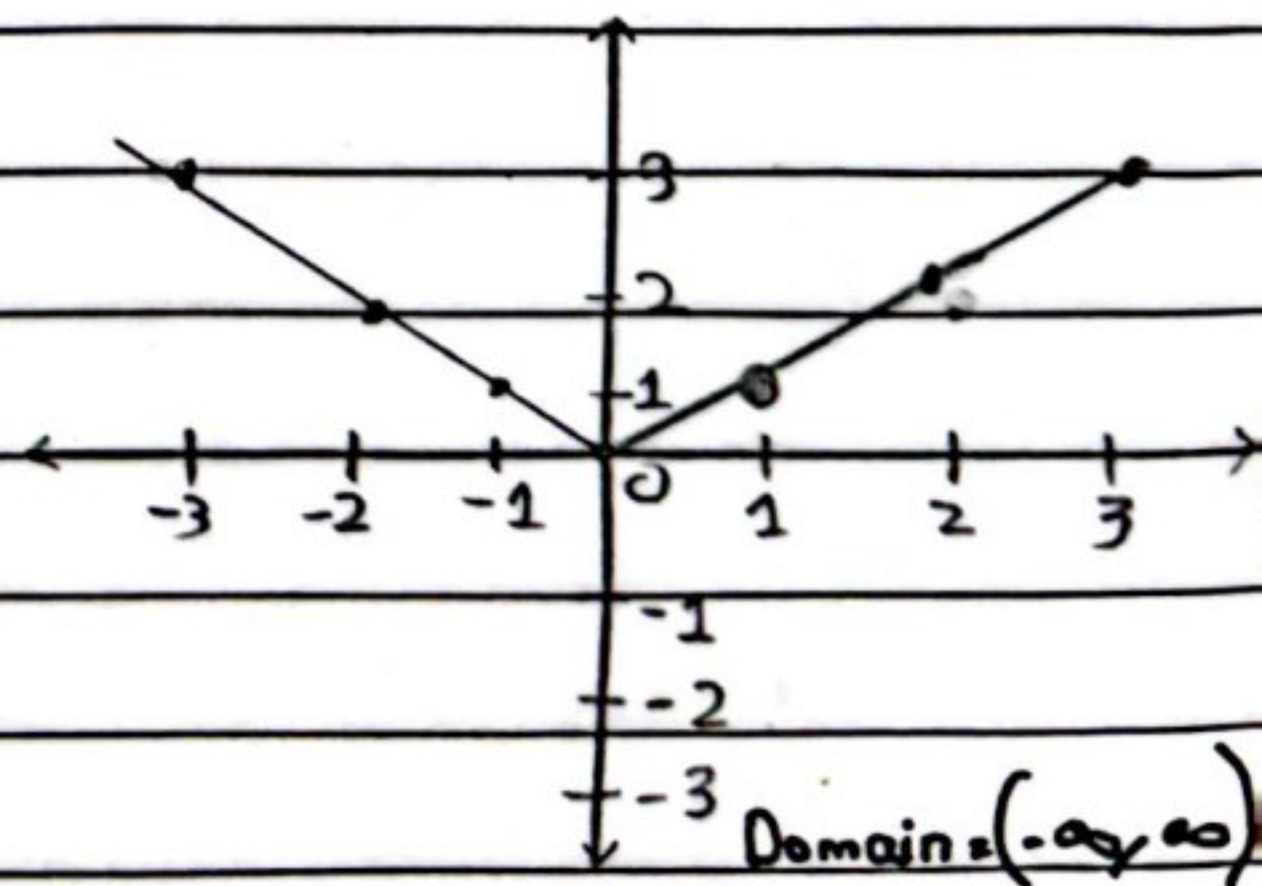
$$|-5| = 5, |3| = 3, |-1/2| = 1/2$$

PROPERTIES OF ABSOLUTE VALUE:-

- $| -a | = | a |$
- $| ab | = | a | | b |$
- $| a/b | = | a | / | b |, b \neq 0$
- $| a+b | \leq | a | + | b |$, triangle inequality

$$y = |x|$$

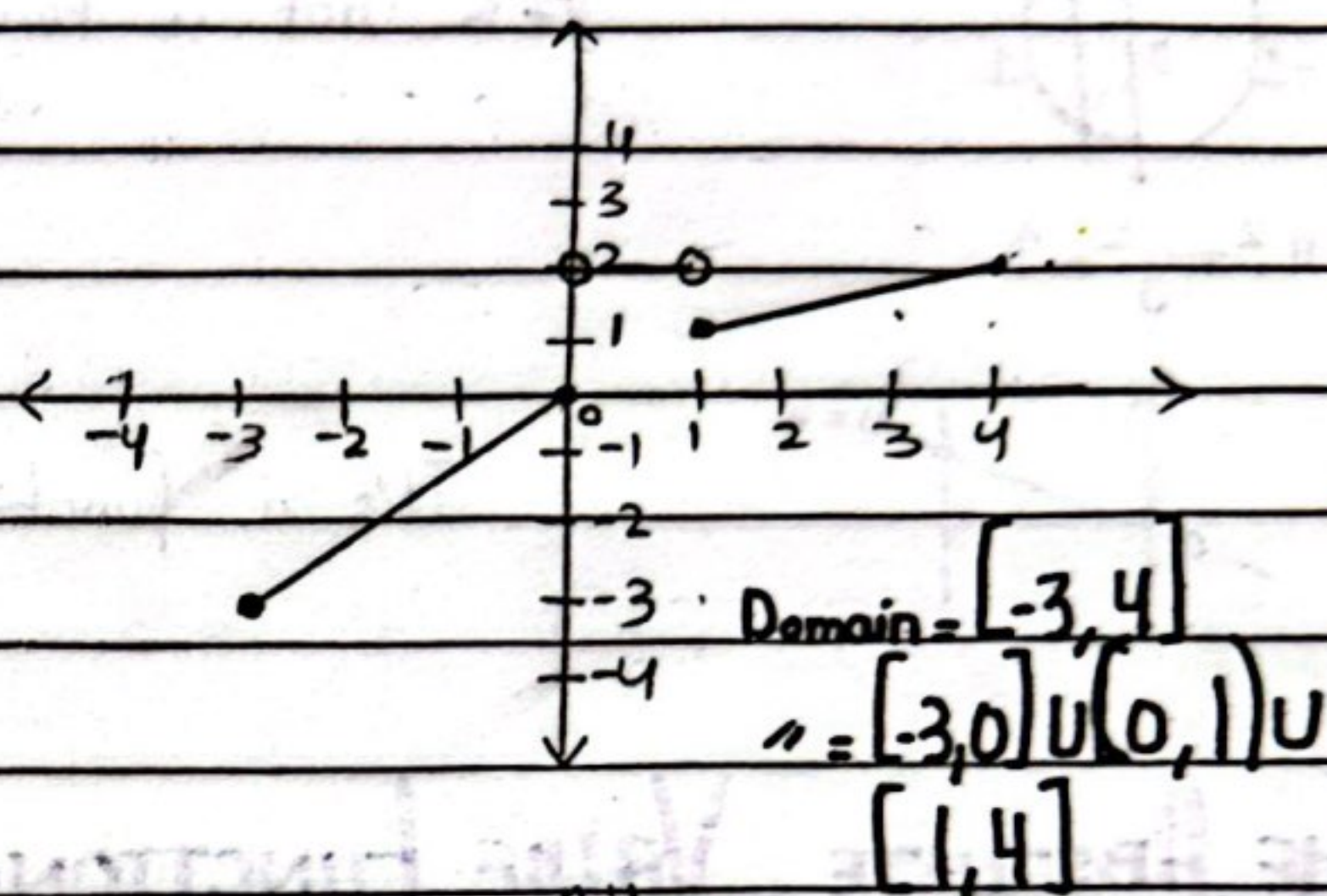
x	-3	-2	-1	0	1	2	3
y = x	3	2	1	0	1	2	3



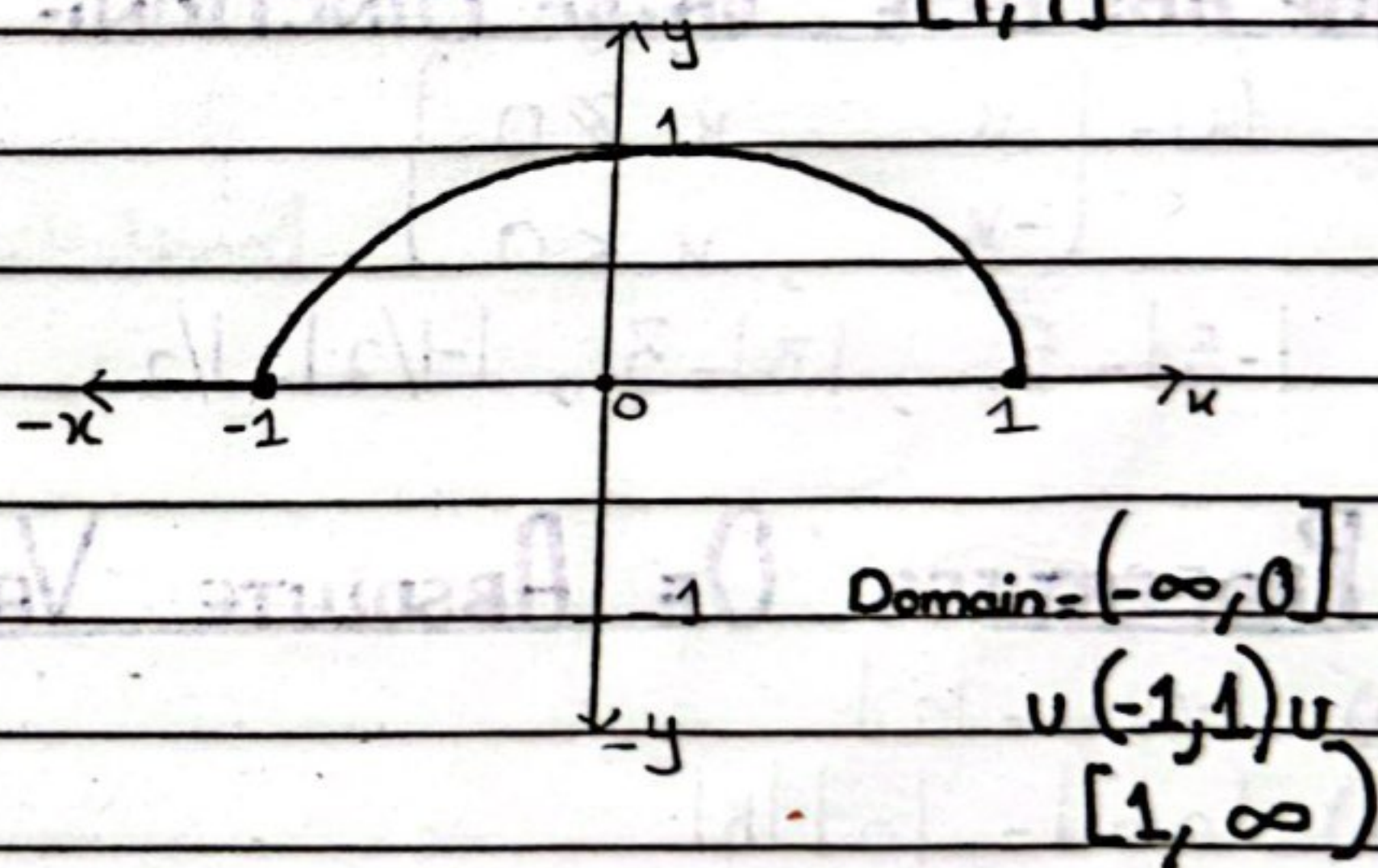
PIECEWISE DEFINED FUNCTION:-

A sketch the graph of the function defined piecewise by the formulae.

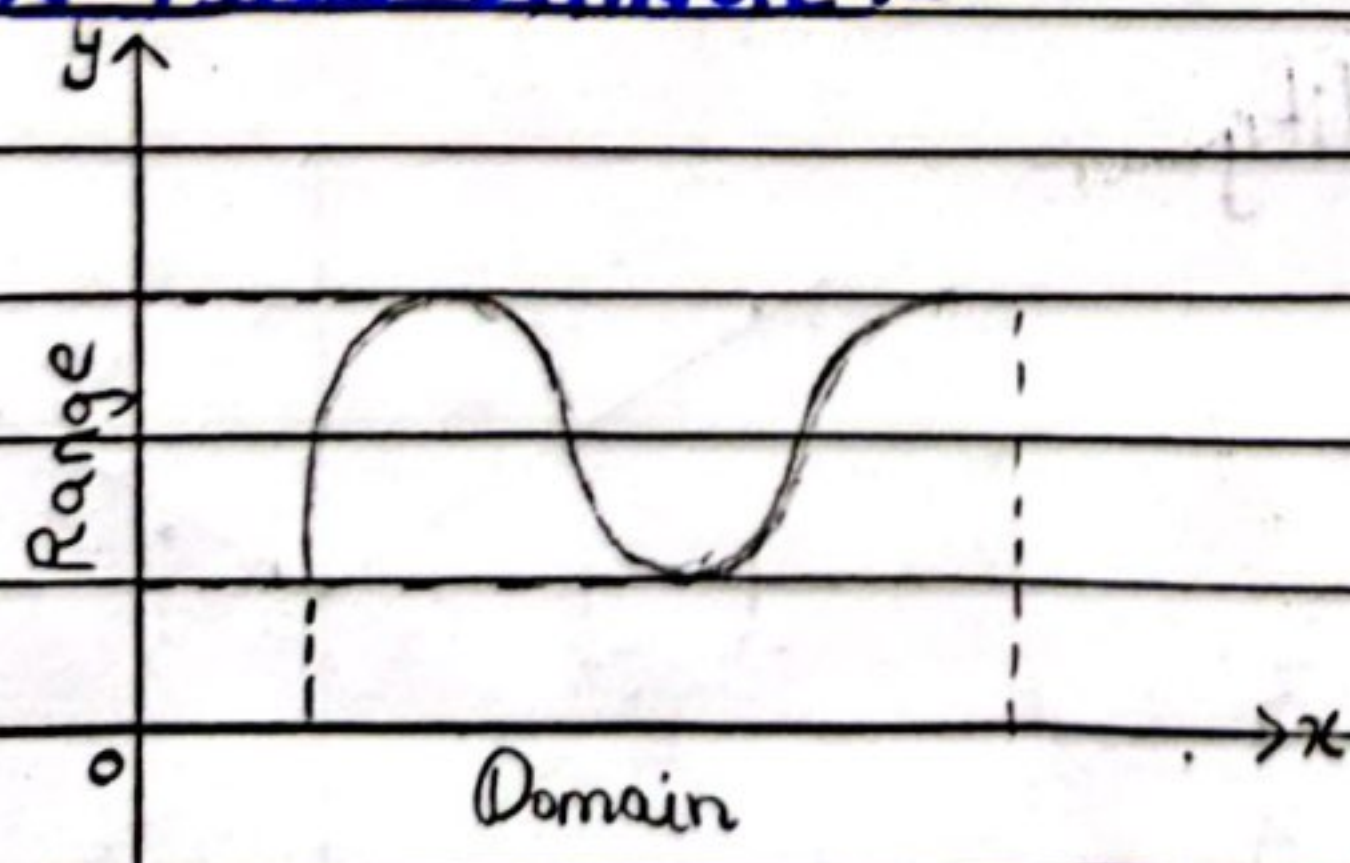
$$f(x) = \begin{cases} x, & -3 \leq x \leq 0 \\ 2, & 0 < x < 1 \\ \sqrt{x}, & 1 \leq x \leq 4 \end{cases}$$



$$Q. f(x) = \begin{cases} 0 & x \leq -1 \\ \sqrt{1-x^2} & -1 < x < 1 \\ x & x \geq 1 \end{cases}$$



DOMAIN AND RANGE:-



R.W

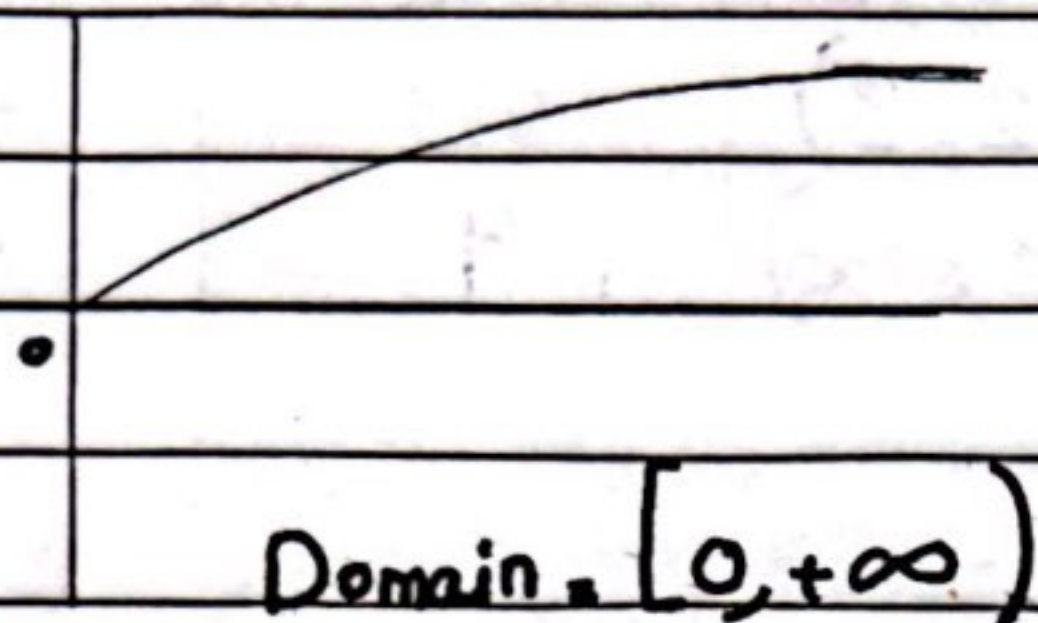
a) $y = 1/x, x \neq 0$

$-\infty \quad 0 \quad \infty$

Domain = $(-\infty, 0) \cup (0, +\infty)$

$$b) y = \sqrt{x}$$

x will be non negative



Q Find the natural domain of

$$a) f(x) = x^3$$

$$(-\infty, +\infty)$$

$$b) f(x) = \frac{1}{(x-1)(x-3)}$$

$$x=1$$

$$x=3$$

$$x \neq 1$$

$$x \neq 3$$

$$\text{Domain} = (-\infty, 1) \cup (1, 3) \cup (3, +\infty)$$

$$c) f(x) = \tan x$$

$$\tan x = \frac{\sin x}{\cos x}, \cos x \neq 0$$

Thus, the natural domain consists of all real numbers except

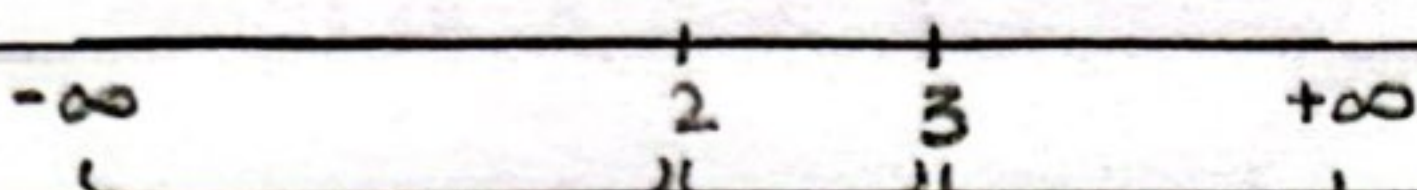
$$x = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots \rightarrow \text{odd integer multiple of } \frac{\pi}{2}$$

$$d) f(x) = \sqrt{x^2 - 5x + 6}$$

$$x^2 - 5x + 6 \geq 0$$

$$(x-2)(x-3) \geq 0$$

Intervals	Test Point	Signs
$(-\infty, 2)$	1	+
$(2, 3)$	2.5	-
$(3, +\infty)$	4	+



$$\text{Domain} = (-\infty, 2] \cup [3, +\infty)$$

ARITHMETIC OPERATIONS ON FUNCTIONS:

$$(f+g)(x) = f(x) + g(x)$$

$$(f-g)(x) = f(x) - g(x)$$

$$(fg)(x) = f(x)g(x)$$

$$(f/g)(x) = f(x)/g(x)$$

Domain defines by intersections of the domains of f & g

Domain defines by intersections of f & g but with the point where $g(x) = 0$, excluded ~~and~~

Q Let $f(x) = 1 + \sqrt{x-2}$ and $g(x) = x-3$. Find the domains and formulas for the functions $f+g$, $f-g$, fg and f/g

$$\begin{aligned} \text{sol } (f+g)(x) &= f(x) + g(x) \\ &= (1 + \sqrt{x-2}) + (x-3) \\ &= x-2 + \sqrt{x-2} \end{aligned}$$

$$\begin{aligned} (f-g)(x) &= (1 + \sqrt{x-2}) - (x-3) \\ &= 4 - x + \sqrt{x-2} \end{aligned}$$

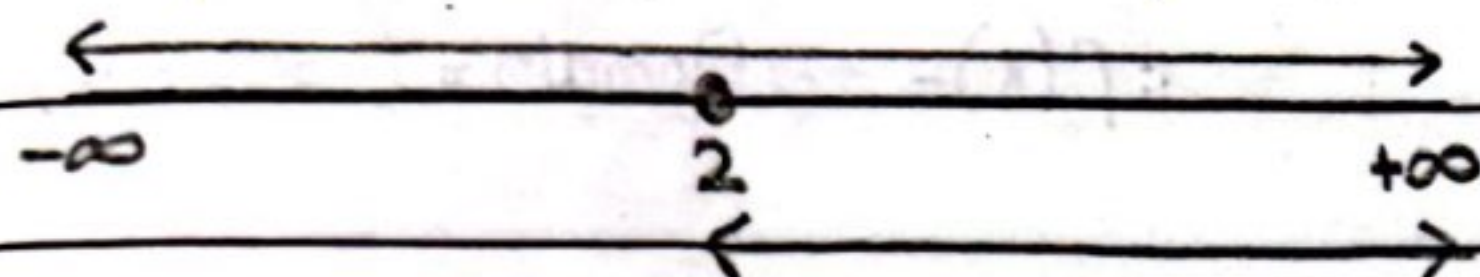
$$fg(x) = (1 + \sqrt{x-2})(x-3)$$

$$f/g(x) = \frac{(1 + \sqrt{x-2})}{x-3}$$

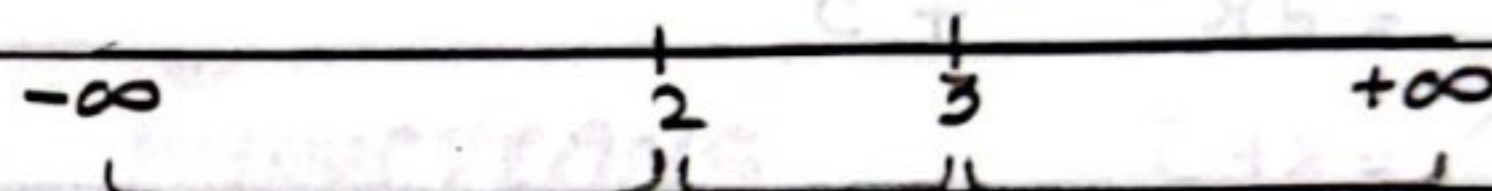
Domain of $f(x) = [2, +\infty)$

Domain of $g(x) = (-\infty, \infty)$

Domain of $(f+g)(x), f-g(x), f \cdot g(x)$ will be the intersections domains of f & g
 $[2, +\infty) \cap (-\infty, +\infty) = [2, +\infty)$ — ①



Since $g(x) = 0$, if $x = 3$, the domain of f/g will be excluding $x = 3$



Domain = $[2, 3) \cup (3, +\infty)$

Q Show that if $f(x) = \sqrt{x}$, $g(x) = \sqrt{x}$ and $h(x) = x$ then the domain of fg is not the same as the natural domain of h .

Domain of $f = [0, +\infty)$

Domain of $g = [0, +\infty)$

Domain of $h = (-\infty, +\infty)$

Domain of $fg = [0, +\infty) \cap [0, +\infty)$
 $= [0, +\infty)$

COMPOSITION OF FUNCTIONS:-

$$(f \circ g)(x) = f(g(x))$$

Q Let $f(x) = x^2 + 3$ and $g(x) = \sqrt{x}$ Find

a) $(f \circ g)(x)$

$\therefore f(x) \rightarrow \text{Domain} = (-\infty, +\infty)$

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(\sqrt{x}) \\ &= (\sqrt{x})^2 + 3 \end{aligned}$$

$$\begin{aligned} (f \circ g)(x) &= x + 3 \\ \text{Domain} &= [0, +\infty) \end{aligned}$$

b) $(g \circ f)(x)$

$$\begin{aligned} (g \circ f)(x) &= g(f(x)) \\ &= g(x^2 + 3) \\ &= \sqrt{x^2 + 3} \end{aligned}$$

Domain of $(g \circ f)(x)$ will be $(-\infty, +\infty)$

EXPRESSING A FUNCTION AS A COMPOSITION:-

$$h(x) = (x+1)^2$$

$$g(x) = x+1, \quad f(x) = x^2$$

$$h(x) = (x+1)^2$$

$$h(x) = [g(x)]^2$$

$$h(x) = f(g(x))$$

$$h(x) = f \circ g(x)$$

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Q Express $\sin(x^3)$ as a Composition of two functions.

$$f(x) = \sin x$$

$$g(x) = x^3$$

$$\begin{aligned}\sin(x^3) &= f(g(x)) \\ &= f(x^3) \\ &= \sin(x^3)\end{aligned}$$

$$\begin{aligned}\frac{1}{x} \\ f \circ g(x) &= f(x+1) \\ &= \frac{1}{x+1}\end{aligned}$$

$$\begin{aligned}f \circ g(x) &= f(4-3x) \\ &= \sqrt{4-3x} \\ f \circ g(x) &= f(\sqrt{x}) \\ &= 8+\sqrt{x}\end{aligned}$$

COMPOSING FUNCTIONS

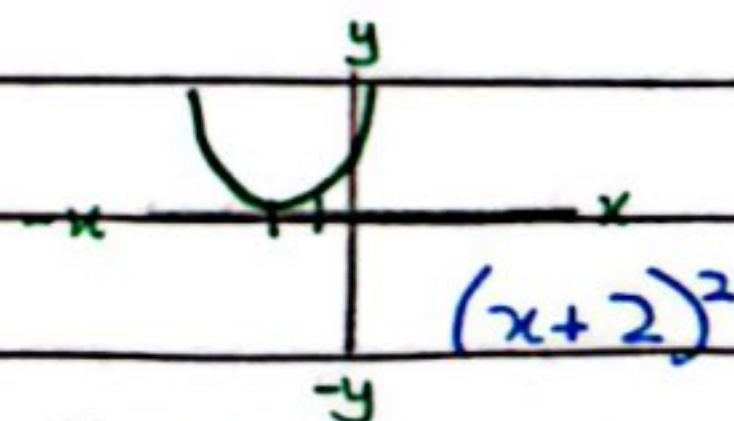
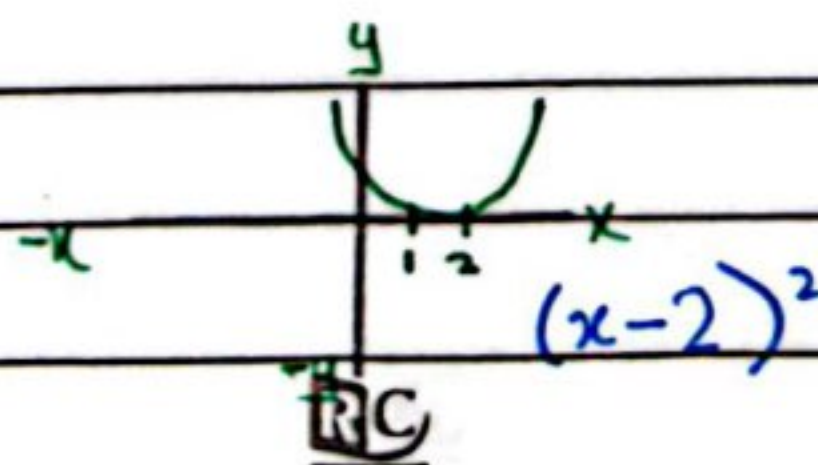
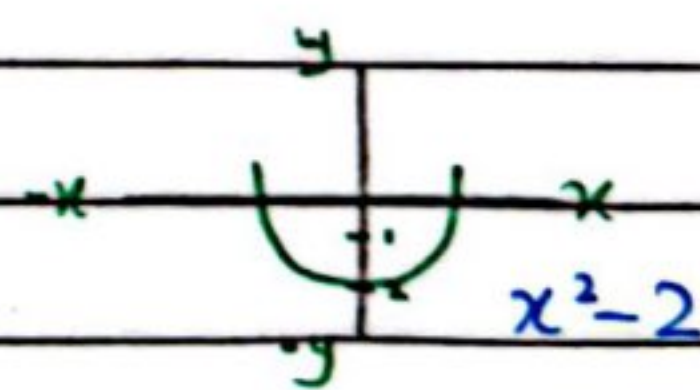
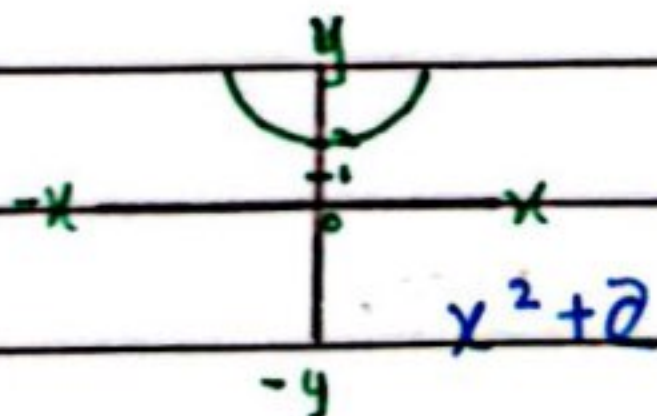
FUNCTIONS	$g(x)$ INSIDE	$f(x)$ OUTSIDE	COMPOSITION
$(x^2+1)^{10}$	x^2+1	x^{10}	$f(g(x))$
$\sin^3 x$	$\sin x$	x^3	$''$
$\tan(x^5)$	x^5	$\tan x$	$''$
$\sqrt{4-3x}$	$4-3x$	$\sqrt{x}$	$''$
$8+\sqrt{x}$	$\sqrt{x}$	$8+x$	$''$
$1/(x+1)$	$x+1$	$1/x$	$''$

$$y = f(x) + c \rightarrow y = x^2 + 2$$

$$y = f(x) - c \rightarrow y = x^2 - 2$$

$$y = f(x+c) \rightarrow y = (x+2)^2$$

$$y = f(x-c) \rightarrow y = (x-2)^2$$



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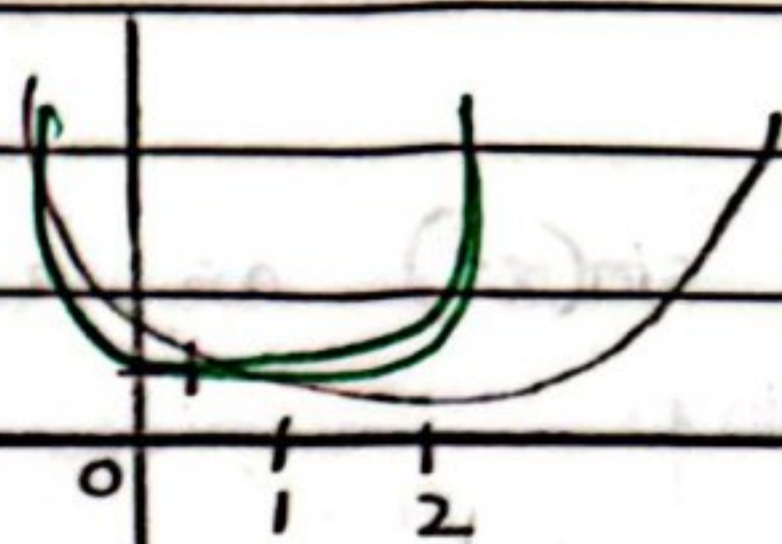
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Q $f(x) = x^2 - 4x + 5$

sol $f(x) = x^2 - 4x + 4 - 4 + 5$

$$= (x^2 - 4x + 4) + 1$$

$$= (x - 2)^2 + 1$$



EXAMPLE 7 :-

(a) $y = \sqrt{x-3}$ (b) $y = \sqrt{x+3}$

Sketch the graph



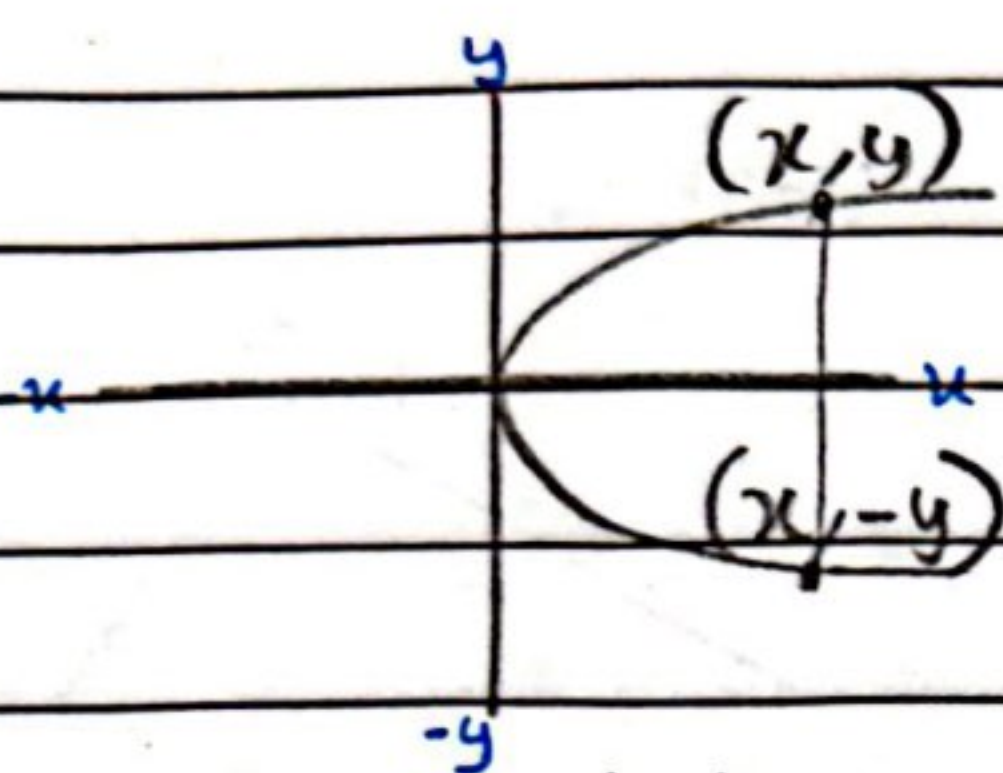
$$\sqrt{x-3}$$

$$\sqrt{x+3}$$

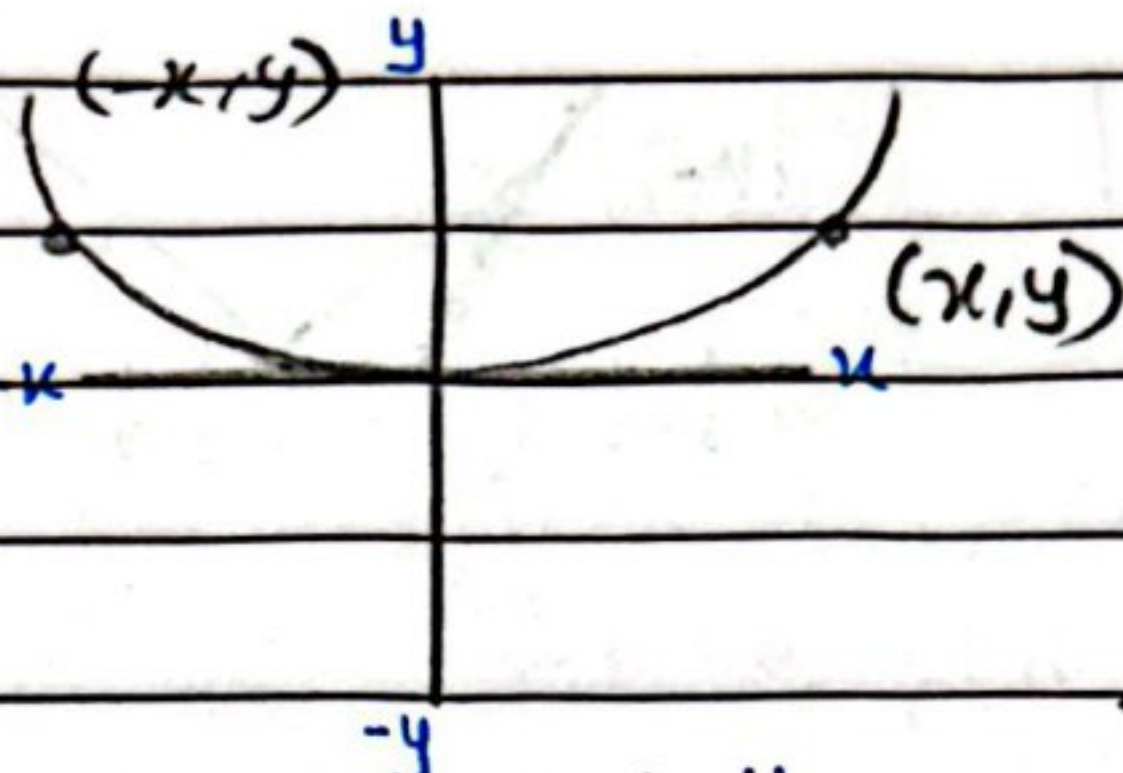
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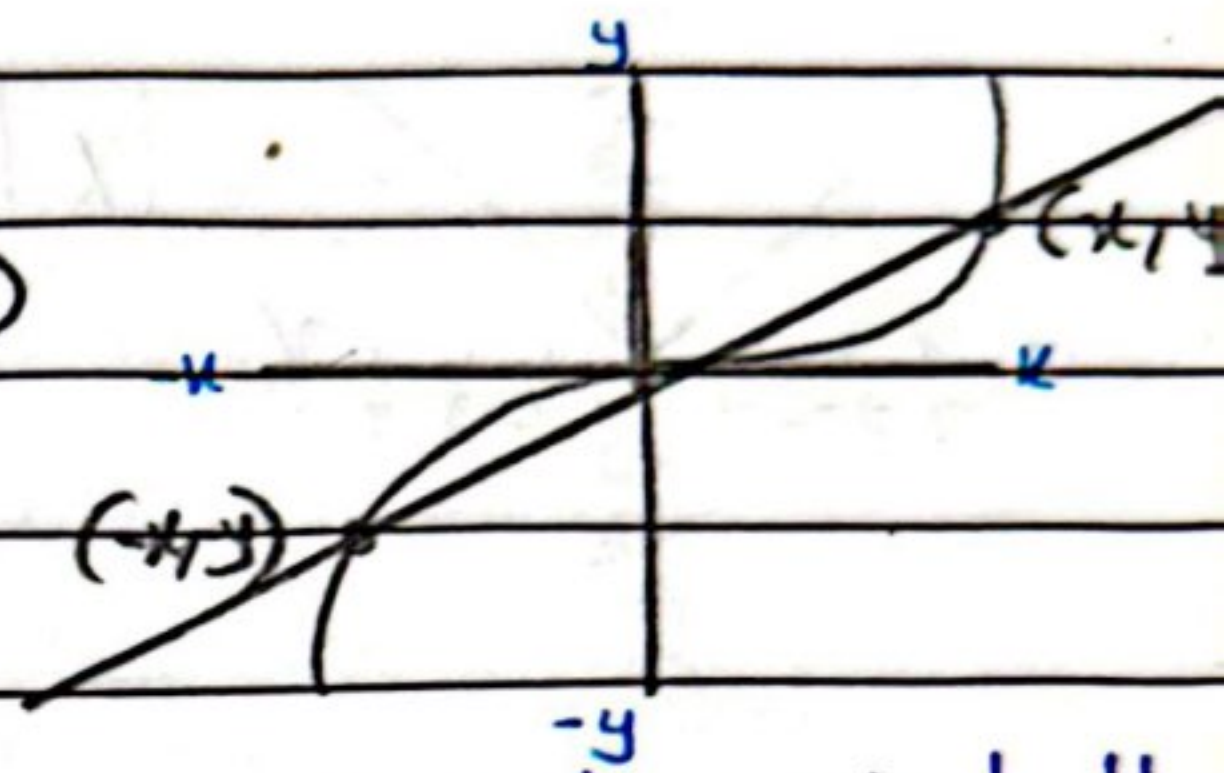
SYMMETRY:-



Symmetry about the
x-axis



Symmetry about the
y-axis

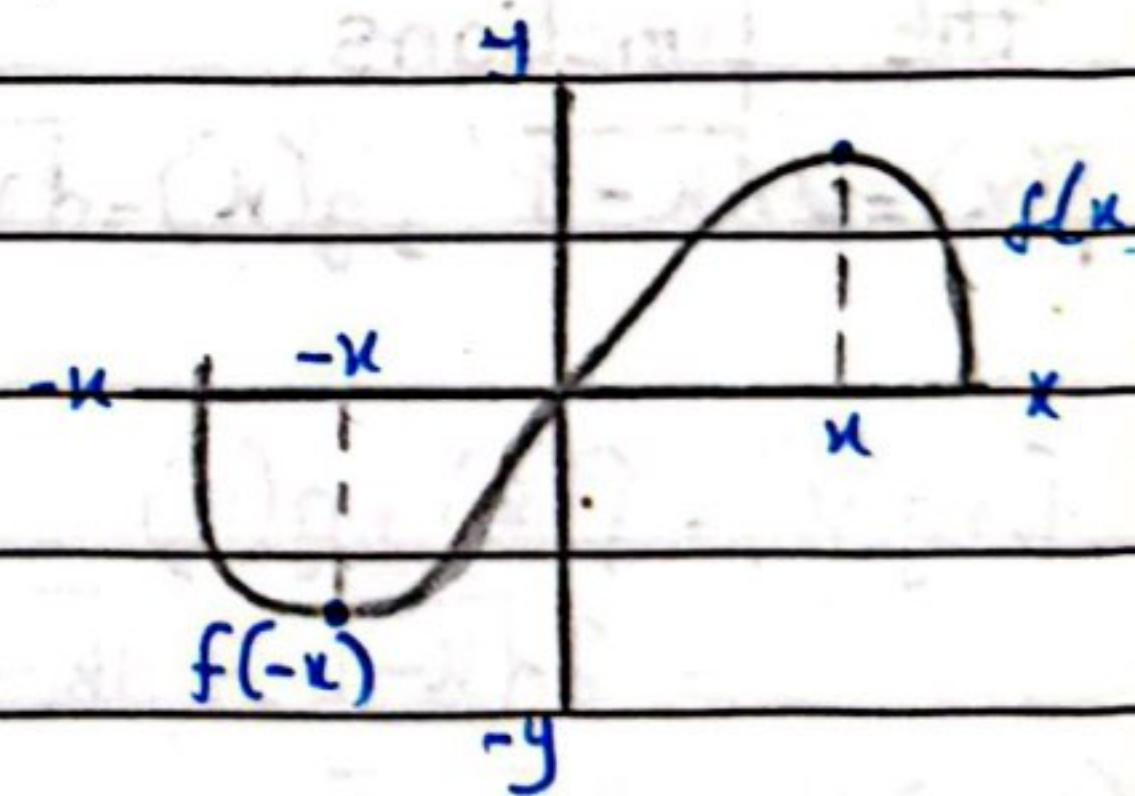
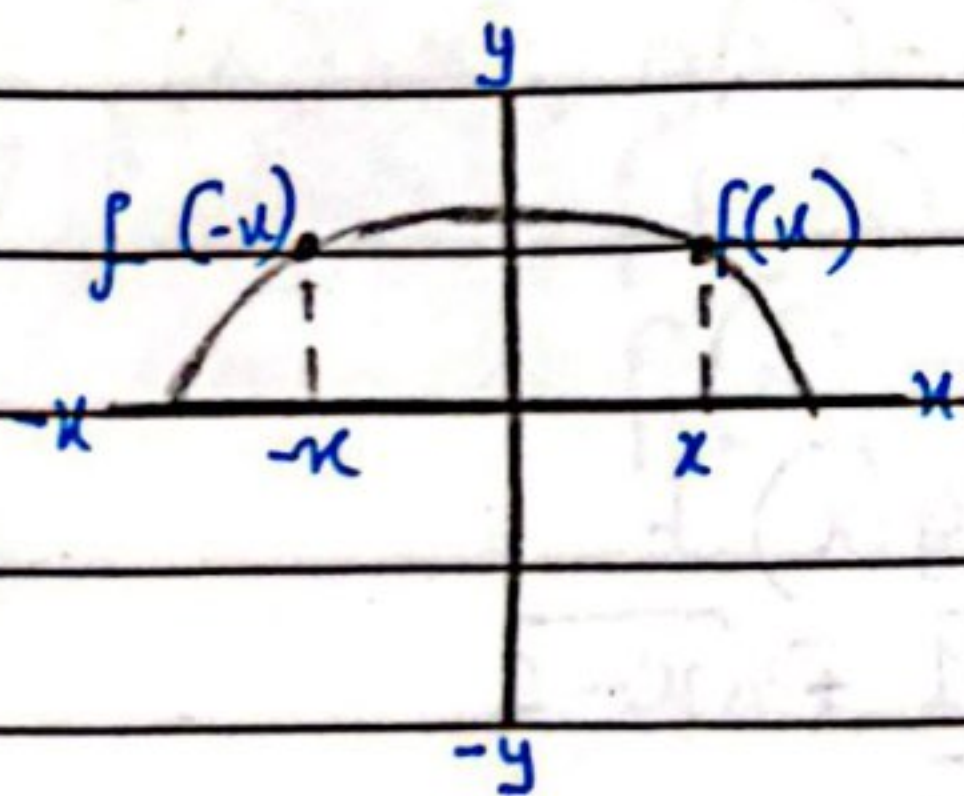


Symmetry about the
origin.

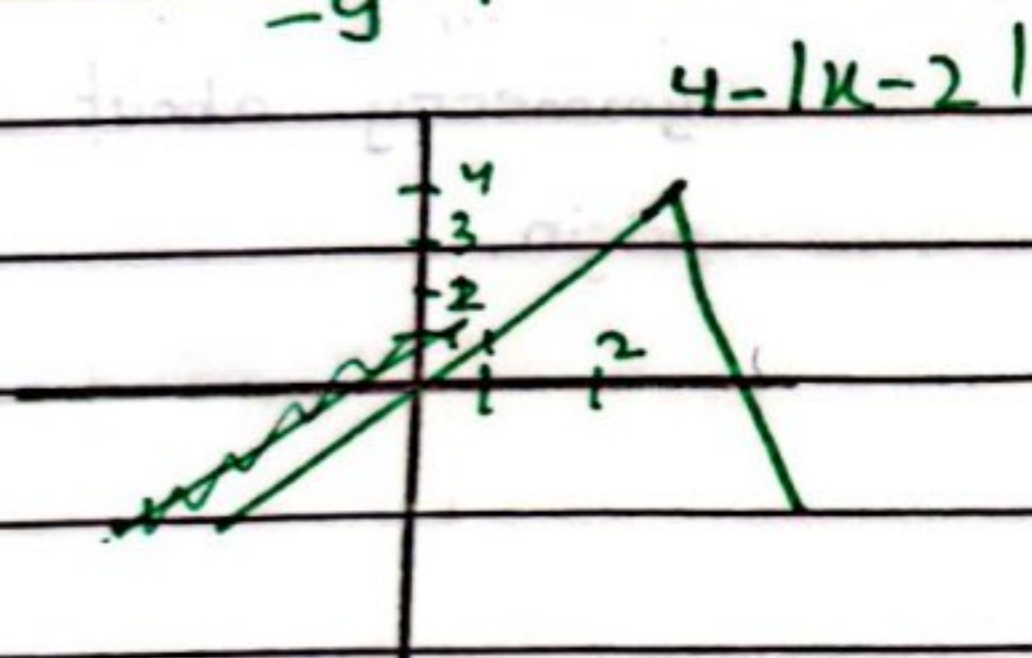
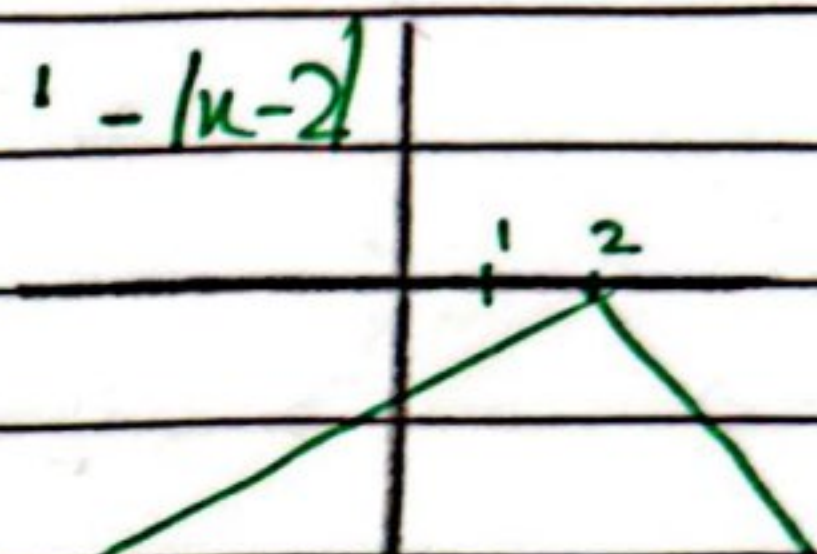
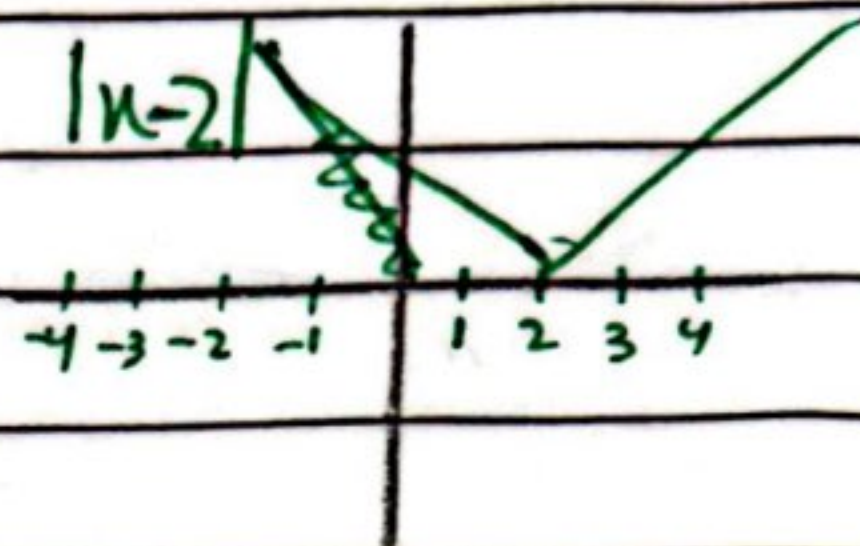
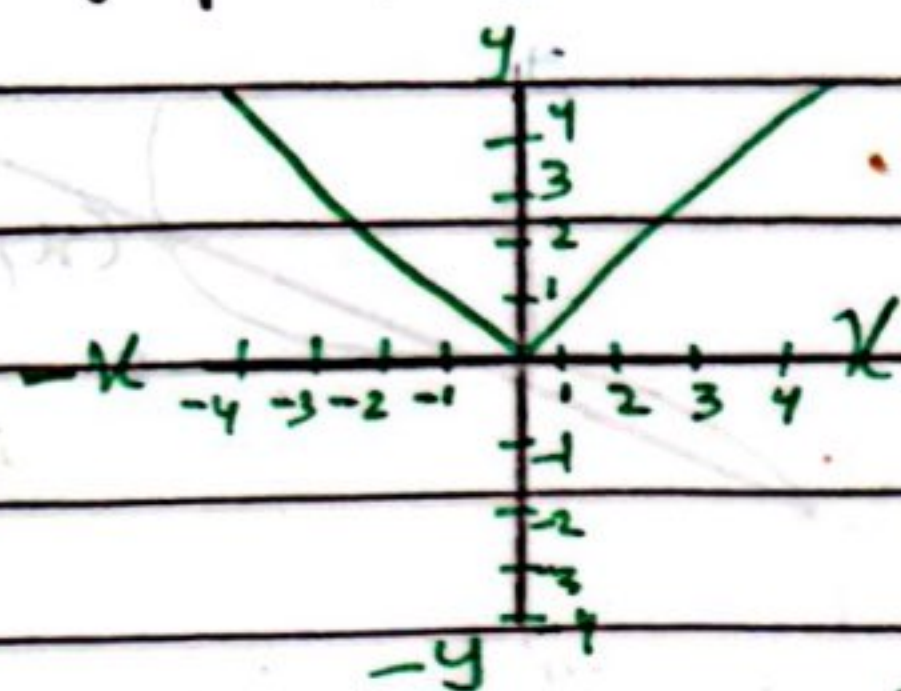
EVEN & ODD FUNCTION:-

$f(-x) = f(x)$, even function

$f(-x) = -f(x)$, odd function



Q:- graph of $y = 4/|x-2|$



27-28 Find formulas for $f+g$, $f-g$, fg , and f/g , and state the domains of the functions.

27. $f(x) = 2\sqrt{x-1}$, $g(x) = \sqrt{x-1}$

sol $f+g(x) = f(x) + g(x)$
 $= 2\sqrt{x-1} + \sqrt{x-1}$

$(f+g)(x) = 3\sqrt{x-1}$

$(f-g)(x) = f(x) - g(x)$
 $= 2\sqrt{x-1} - \sqrt{x-1}$

$(f-g)(x) = \sqrt{x-1}$

$(fg)(x) = (2\sqrt{x-1})(\sqrt{x-1})$

$(f/g)(x) = \frac{2\sqrt{x-1}}{\sqrt{x-1}}$

Domain: $f(x) \rightarrow [1, +\infty)$
 $g(x) \rightarrow [1, +\infty)$

If $g(x) = 0$ and $f(x) = 1$ f/g will be excluding 1.

Domain of fg , $f+g$, $f-g$ will be the intersections domains of f & g $[1, +\infty) \cap [1, +\infty) = [1, +\infty)$

28. $f(x) = \frac{x}{1+x^2}$, $g(x) = \frac{1}{x}$

sol $(f+g)(x) = \left(\frac{x}{1+x^2}\right) + \left(\frac{1}{x}\right)$

$(f-g)(x) = \left(\frac{x}{1+x^2}\right) - \left(\frac{1}{x}\right)$

$(fg)(x) = \left(\frac{x}{1+x^2}\right) \left(\frac{1}{x}\right)$

$(f/g)(x) = \frac{x}{1+x^2} \div \frac{1}{x}$

Domain: $f(x) = [1, \infty)$
 $g(x) = [1, \infty)$

Since $g(x) = 0$ $x =$

Domain $f(x) = (-\infty, \infty)$

$g(x) = (-\infty, 0) \cup (0, \infty)$

Intersection: $(-\infty, \infty) \cap (-\infty, 0) \cup (0, \infty)$
 $\therefore (-\infty, 0) \cup (0, \infty)$

31-34 Find formulas for $f \circ g$ and $g \circ f$, and state the domains of the comp

31. $f(x) = x^2$, $g(x) = \sqrt{1-x}$

sol $(f \circ g)(x) = f(g(x))$
 $= f(\sqrt{1-x})$
 $= (\sqrt{1-x})^2$

$= 1-x$

$= (-\infty, 1]$

$(g \circ f)(x) = g(f(x))$

$= g(x^2)$

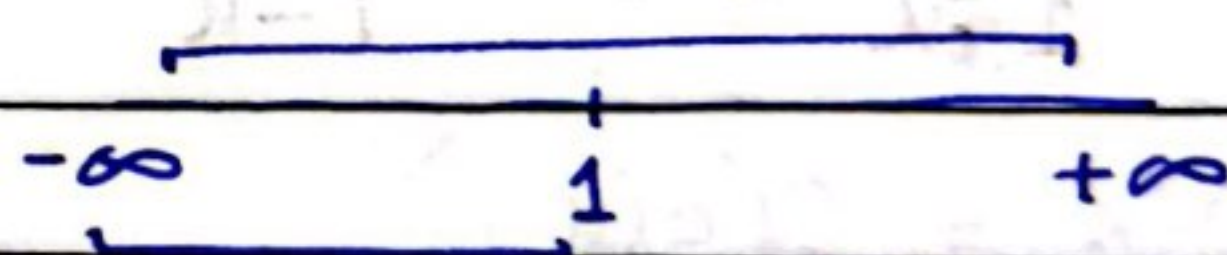
$= \sqrt{1-x^2}$

$= [-1, +1]$

$\Rightarrow 1-x^2 \geq 0 \Rightarrow 1 \geq x^2$

$f(x) = (-\infty, +\infty)$

$g(x) = [-\infty, +1]$



32. $f(x) = \sqrt{x-3}$, $g(x) = \sqrt{x^2+3}$

Domain: $f(x) = x-3 \geq 0 \Rightarrow x \geq 3 \Rightarrow [3, +\infty)$

Sol $(f \circ g)(x) = f(g(x))$
 $= f(\sqrt{x^2+3})$
 $= \sqrt{\sqrt{x^2+3}-3}$
 $= \sqrt{x^2+3-3}$

$g(x) = \sqrt{x^2+3}$
 $-x^2+3 \geq 0 \Rightarrow (-\infty, +\infty)$

$(f \circ g)(x) = \sqrt{x^2+3-3} \Rightarrow \sqrt{x^2+3-3} \geq 0 \Rightarrow \sqrt{x^2+3} \geq 3 \Rightarrow x^2+3 \geq 9 \Rightarrow x^2 \geq 6 \Rightarrow x \geq \sqrt{6}$

$(g \circ f)(x) = g(f(x))$
 $= g(\sqrt{x-3})$
 $= \sqrt{(\sqrt{x-3})^2+3}$
 $= \sqrt{x-3+3}$

$(g \circ f)(x) = \sqrt{x} \Rightarrow x \geq 0$

33. $f(x) = \frac{1+x}{1-x}$, $g(x) = \frac{x}{1-x}$

Sol $(f \circ g)(x) = f(g(x))$
 $= f\left(\frac{x}{1-x}\right)$

Domain: $f(x) = (-1, \infty)$
 $g(x) = [2, +\infty)$

$= \frac{1+\left(\frac{x}{1-x}\right)}{1-\left(\frac{x}{1-x}\right)}$
 $= \frac{(1-x)+x}{1-x}$

$= \frac{(1-x)-x}{1-x}$

$= (-\infty, +1) \cup (+1, \infty)$
 $g(x) = (-\infty, 1) \cup (1, \infty)$

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$$(f \circ g)(x) = \frac{(1-x)+x}{(1-x)-x}$$

$$= \frac{1-x+x}{1-x-x}$$

$$= \frac{1}{1-2x}$$

Domain of fg , $f+g$, $f-g$ will be intersections domains of f & g $[(-\infty, +1) \cup (+1, \infty) \cap (-\infty, +1) \cup (+1, \infty)] = (-\infty, +1) \cup (+1, \infty)$

CONTINUITY:-

A function 'f' is said to be continuous at $x=c$ provided the following condition is satisfied.

- 1) $f(x)$ is defined
- 2) $\lim_{x \rightarrow c} f(x)$ exists.

$$3) \lim_{x \rightarrow c} f(x) = f(c)$$

Ex#1

Determine if whether the following functions are continuous at $x=2$

a) $f(x) = \frac{x^2-4}{x-2}$

sol L.H.L $\Rightarrow$

$$\lim_{x \rightarrow 2^-} \frac{x^2-4}{x-2} = \frac{x^2-4}{x-2} = \frac{(x^2-2^2)}{x-2} \Rightarrow \frac{(x-2)(x+2)}{(x-2)} = 2+2=4$$

R.H.L $\Rightarrow$

$$\lim_{x \rightarrow 2^+} \frac{x^2-4}{x-2} = \frac{x^2-2^2}{x-2} = \frac{(x-2)(x+2)}{(x-2)} = 2+2=4$$

b) $g(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 3, & x = 2 \end{cases}$

c) $h(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$

sol L.H.L $\Rightarrow$

$$\lim_{x \rightarrow 2^-}$$

LIMITS OF FUNCTIONS:-

If the value of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but not equal to a) then we write,

$$\lim_{x \rightarrow a} f(x) = L$$

which is read "the limit of $f(x)$ as x approaches a is L " or " $f(x)$ approaches L as x approaches a ".

Ex# 1.

Use numerical evidence to make a conjecture what the value of $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1}$

x	0.99	0.999	0.9999	0.99999
$f(x)$	1.995	1.9995	1.99995	1.999995

x	1.00001	1.0001	1.001	1.01
$f(x)$	2.000005	2.00005	2.0005	2.005

$$\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1}$$

Sol $\frac{x-1}{\sqrt{x}-1} \times \frac{\sqrt{x}+1}{\sqrt{x}+1}$

$$\frac{(x-1)(\sqrt{x}+1)}{(\sqrt{x}-1)(\sqrt{x}+1)}$$

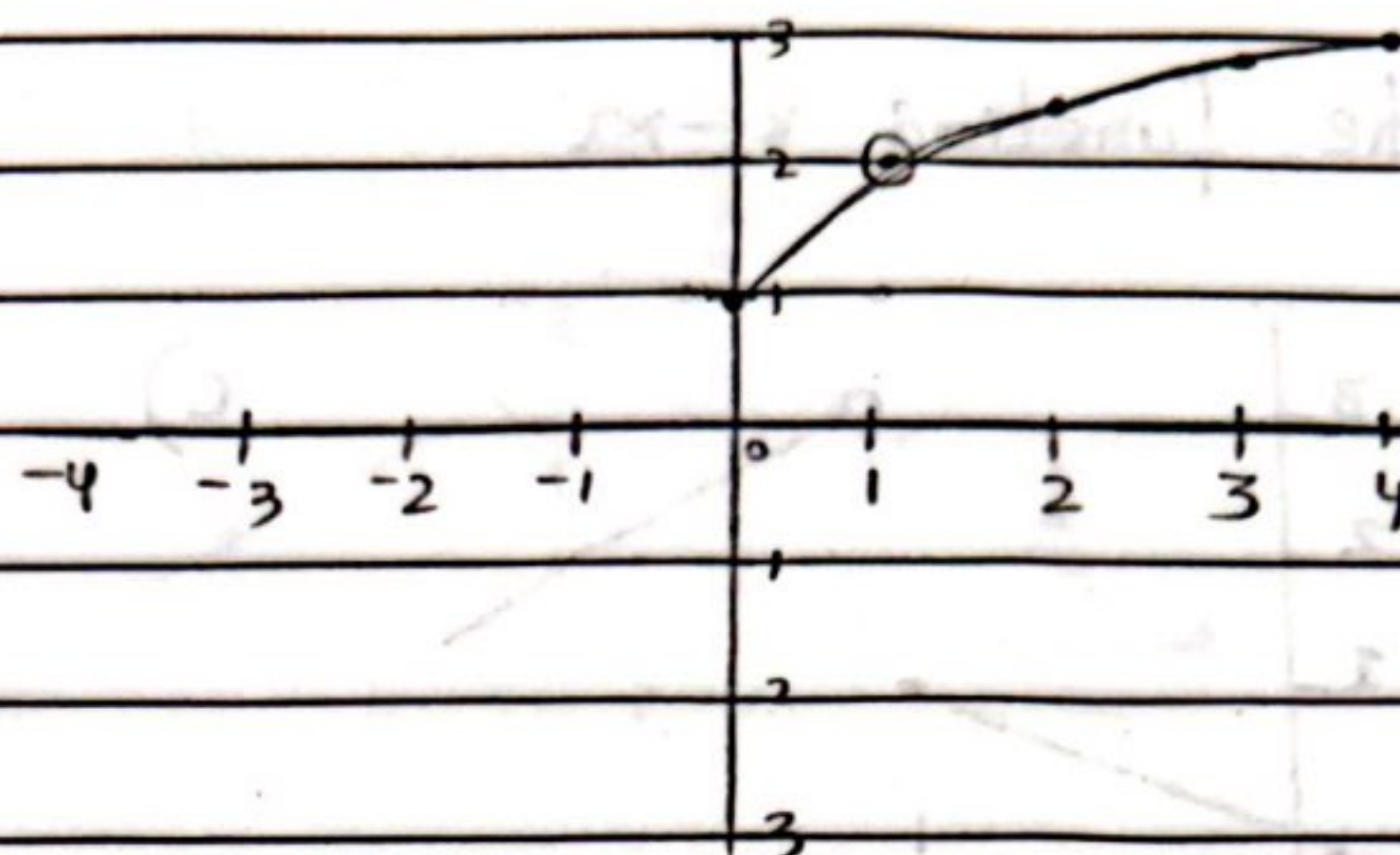
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$$\lim_{x \rightarrow 1} \frac{(x-1)(\sqrt{x}+1)}{(\sqrt{x})^2 - (1)^2}$$

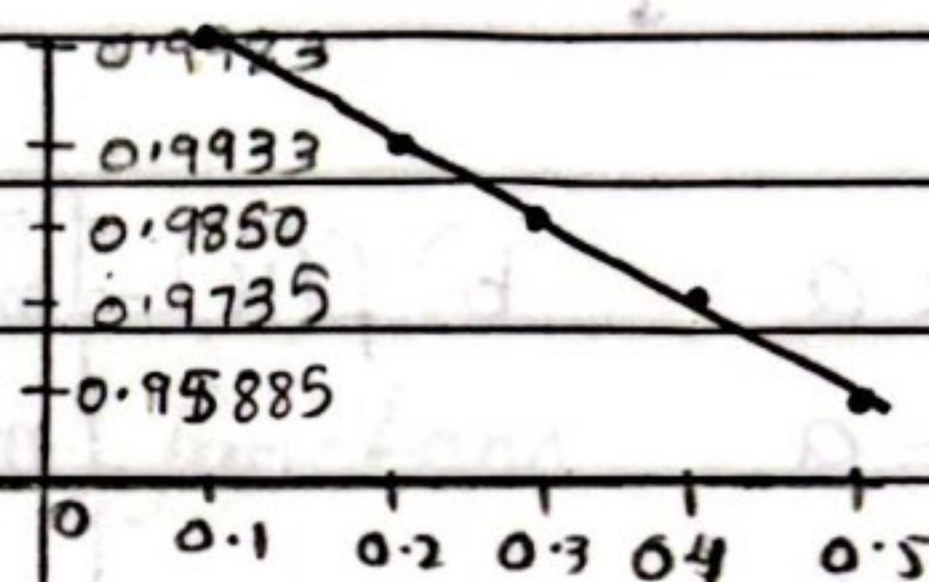
$$\lim_{x \rightarrow 1} \frac{(x-1)(\sqrt{x}+1)}{(x-1)}$$

$$\lim_{x \rightarrow 1} \sqrt{x} + 1$$



x	$f(x)$
0	1
2	2.414
3	2.732
4	3

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$



x	$f(x)$
0.1	0.9983
0.2	0.9933
0.3	0.9850
0.4	0.9735
0.5	0.9588

Ex#3 Find the limit of the following

$$f(x) = \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

At $x=0$

$$\text{L.H.L.} \Rightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{-1}{-1} \Rightarrow -1$$

$$\text{R.H.L.} \Rightarrow \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{1}{1} \Rightarrow 1$$

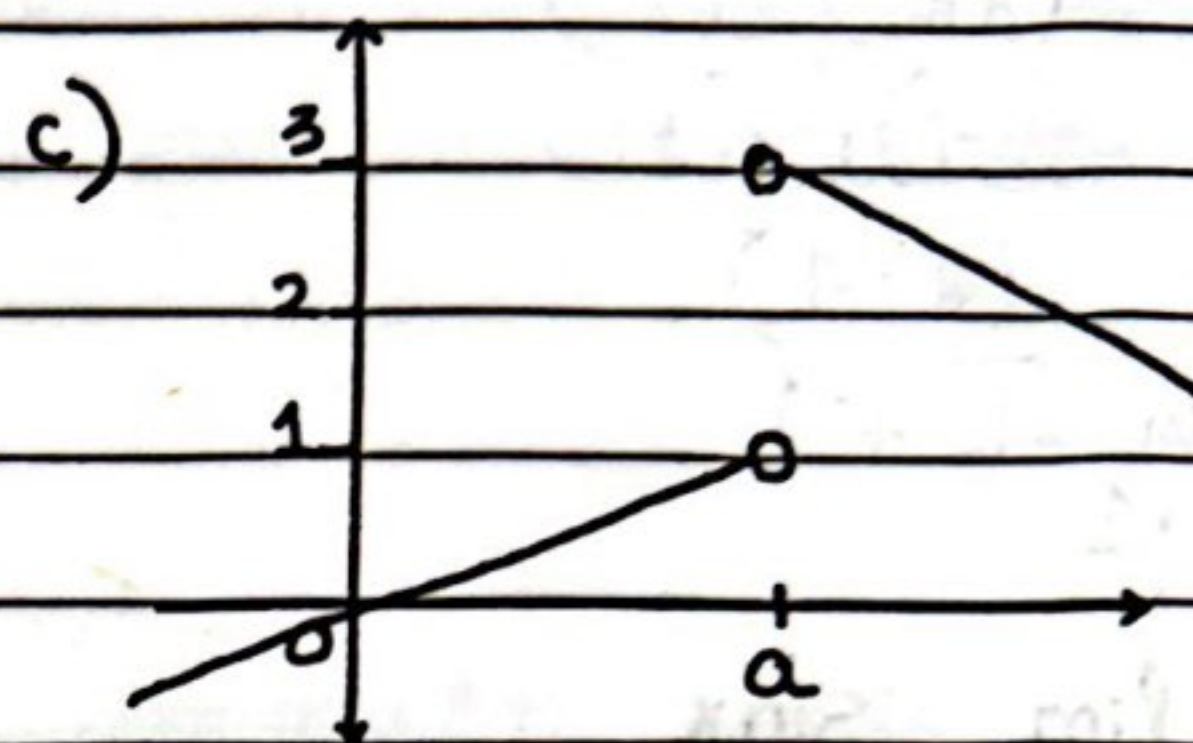
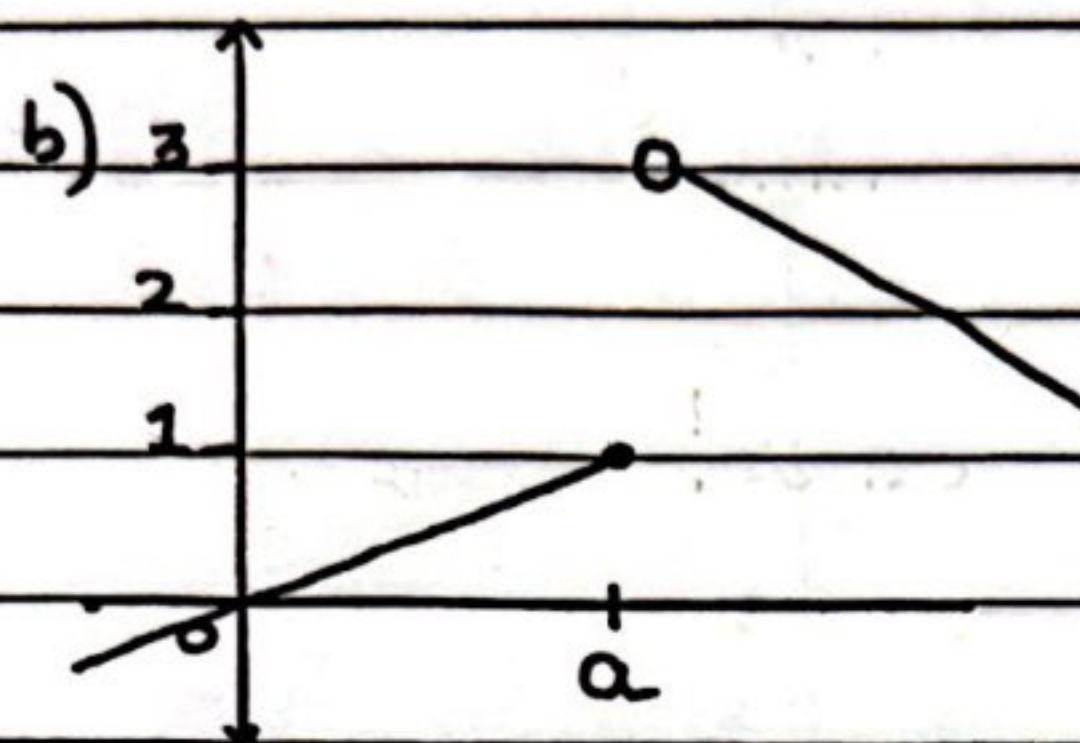
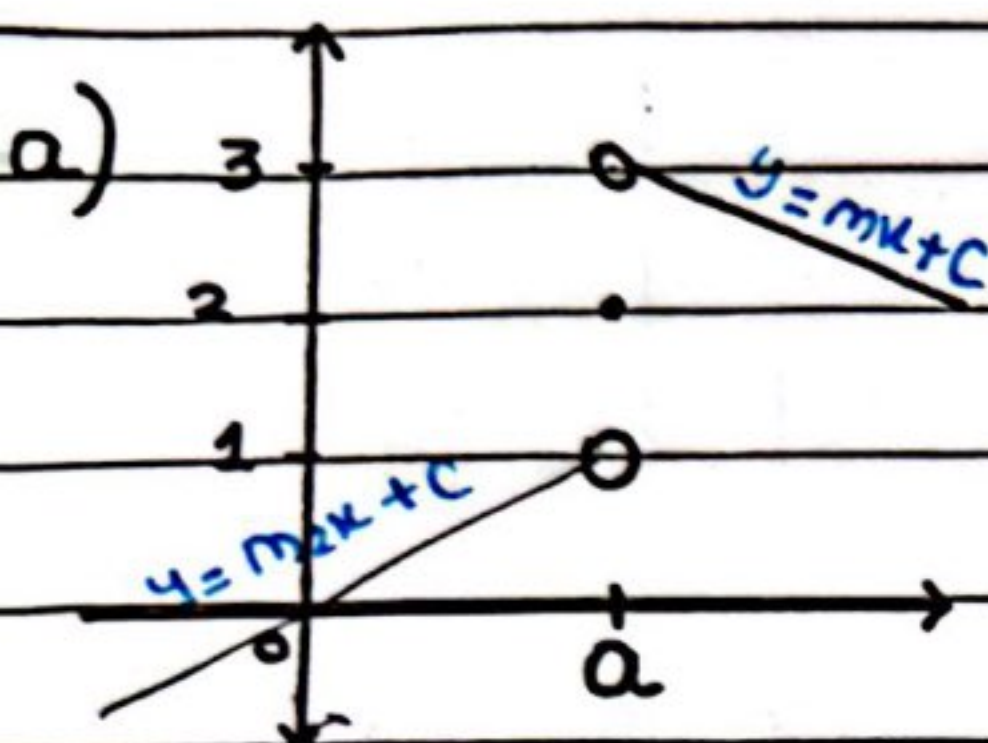
L.H.L. $\neq$ R.H.L.At $x=1$

$$\text{L.H.L.} \Rightarrow \lim_{x \rightarrow 1^-} f(x) = 1$$

L.H.L. = R.H.L.

$$\text{R.H.L.} \Rightarrow \lim_{x \rightarrow 1^+} f(x) = 1$$

Ex#4 Find the limits of the function $x \rightarrow a$

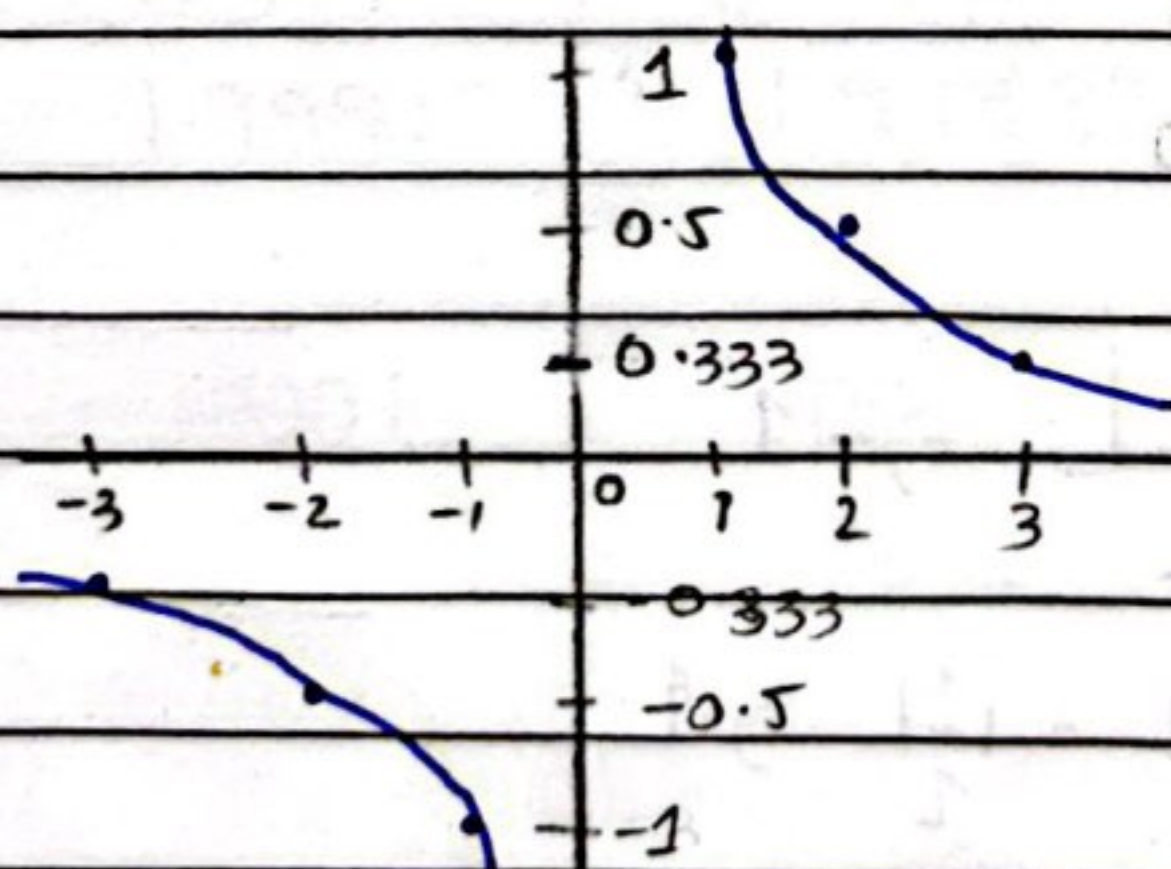


$$a) f(x) = \begin{cases} m_1x+c, & x < a \\ 2, & x = a \\ m_2x+c, & x > a \end{cases}$$

$$b) f(x) = \begin{cases} m_2x+c, & x \leq a \\ m_1x+c, & x > a \end{cases}$$

$$c) f(x) = \begin{cases} m_2x+c, & x < a \\ m_1x+c, & x > a \end{cases}$$

Ex#5 Find limit of the $f(x) = \frac{1}{x}$ where $x \rightarrow 0$

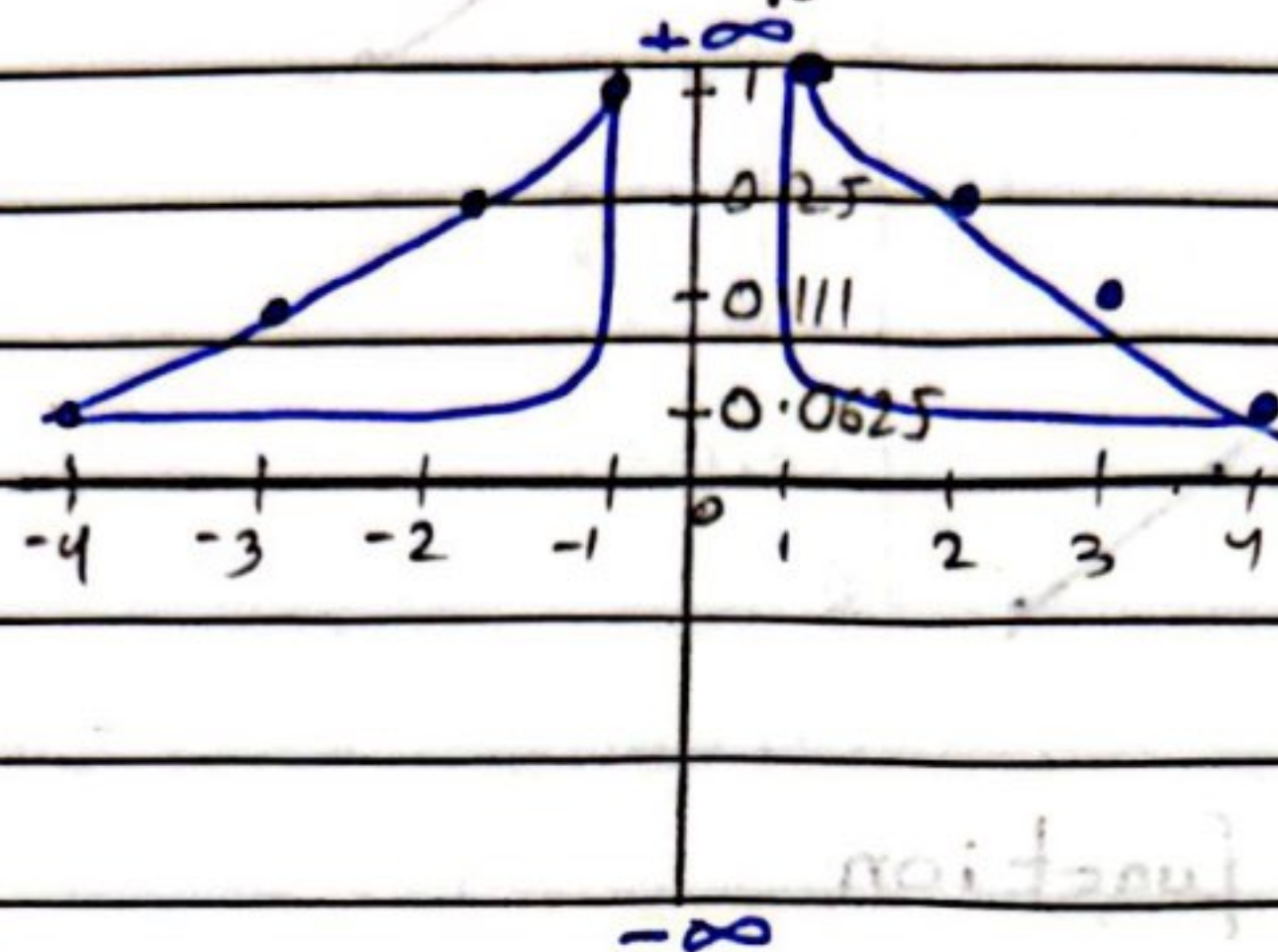


x	$f(x)$
1	1
2	0.5
3	0.333
-1	-1
-2	-0.5
-3	-0.333

$$L.H.L = \lim_{x \rightarrow 0^-} f(x) = -\infty$$

$$R.H.L = \lim_{x \rightarrow 0^+} f(x) = +\infty$$

Ex#6 Find limit of the $f(x) = \frac{1}{x^2}$ where $x \rightarrow 0$

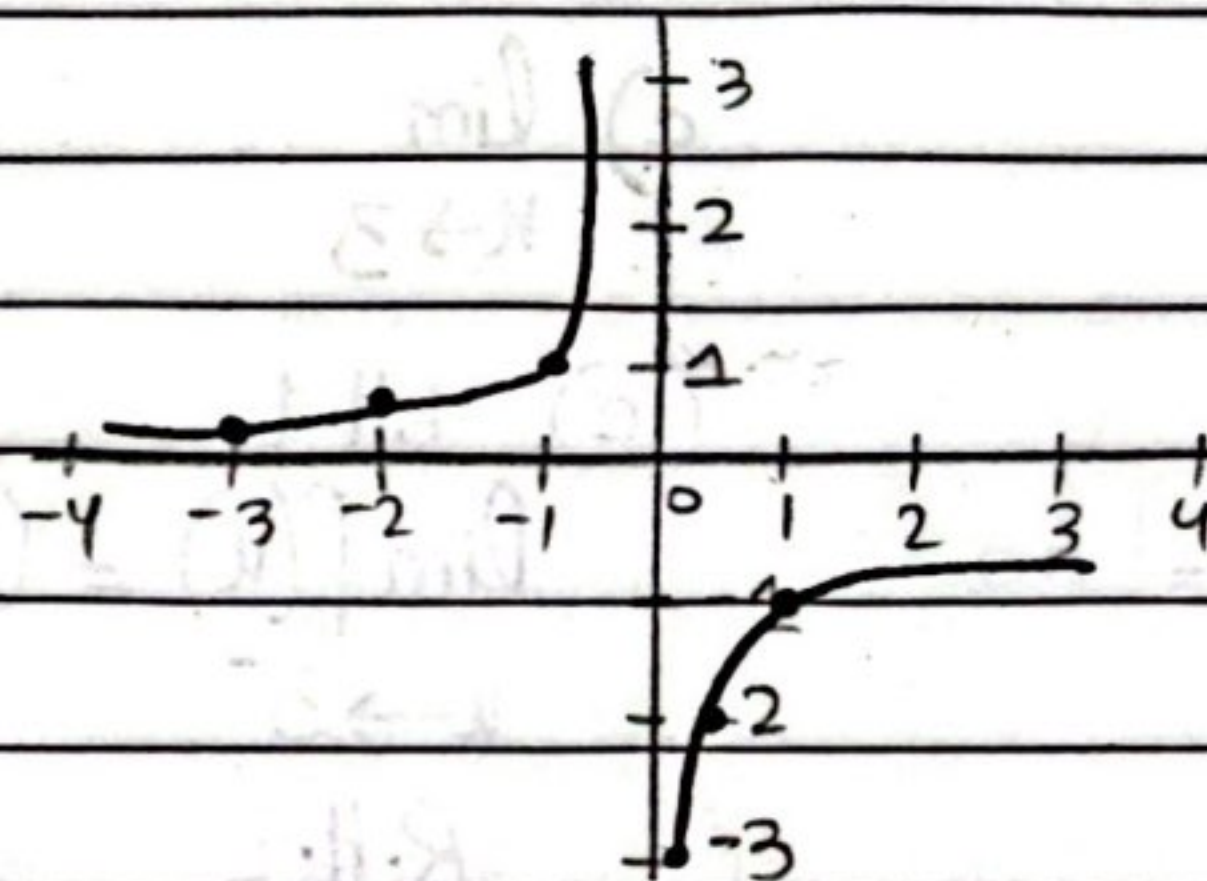


x	$f(x)$
1	1
2	0.25
3	0.111
4	0.0625
-1	1
-2	0.25
-3	0.111
-4	0.0625

Ex# Find the limits of the following functions.

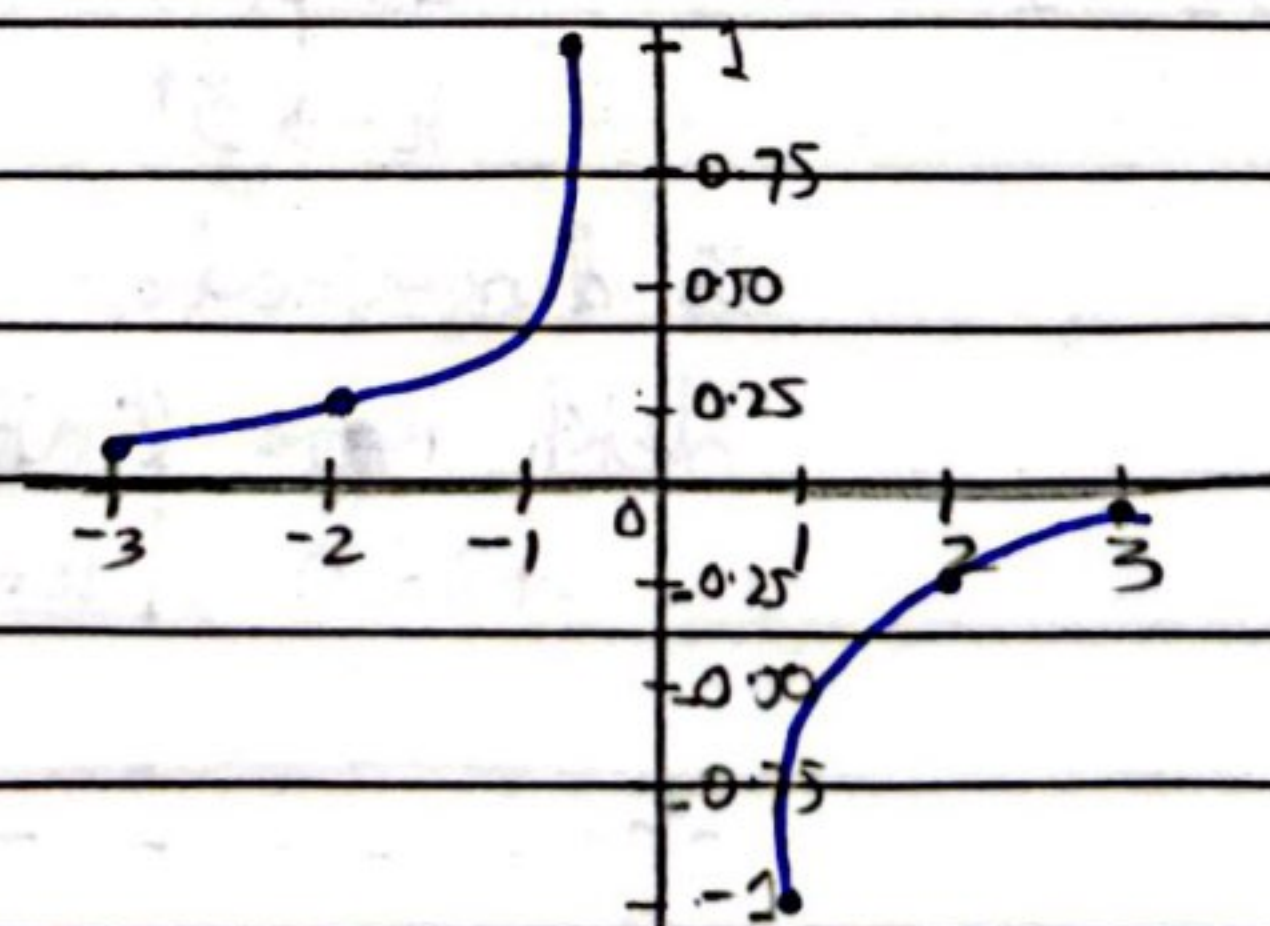
a) $\lim_{x \rightarrow 0} -\frac{1}{x}$

sol



x	$f(x)$
1	-1
2	-0.5
3	-0.33
-1	1
-2	0.5
-3	0.33

b) $\lim_{x \rightarrow 0} -\frac{1}{x^2}$



x	$f(x)$
1	-1
2	-0.25
3	-0.111
-1	-1
-2	-0.25
-3	-0.111

The graph shows the function $y = 1 - |x|$ for $-3 \leq x \leq 3$. The x-axis is labeled from -3 to 3, and the y-axis is labeled from -1 to 1. The function is plotted as a solid black line with dots at the points $(-3, -2)$, $(-2, -1)$, $(-1, 0)$, $(0, 1)$, $(1, 0)$, $(2, -1)$, and $(3, -2)$.

x	$f(x)$
1	1
2	0.125
3	0.037
-1	-1
-2	-0.125
-3	-0.037

$$f(x) = \begin{cases} \frac{1}{x+2}, & x < -2 \\ x^2 - 5, & -2 < x \leq 3 \\ \sqrt{x+13}, & x > 3 \end{cases}$$

$$a) \lim_{n \rightarrow -2} f(n)$$

$$b) \lim_{x \rightarrow 0} f(x)$$

$$c) \lim_{n \rightarrow 3}$$

(c) L.H.L.

$$\lim_{u \rightarrow -2^-} f(u) = \frac{1}{-2+2} = \frac{1}{0} = \infty$$

$$\lim_{x \rightarrow 3^-} f(x) = (3)^2 - 5 = 9 - 5 = 4$$

R.H.L

R.H.L

$$\lim_{x \rightarrow -2^+} f(x) = (-2)^2 - 5 = 4 - 5 = -1$$

$$\lim_{x \rightarrow 3^+} f(x) = \sqrt{(3) + 13} = \sqrt{16} = 4$$

(b)

L.H.L

$$\lim_{x \rightarrow 0} f(x) = (0)^2 - 5 = -5$$

$$K \rightarrow O^-$$

$$R \cdot H \cdot L = (0)^2 - 5 = -5$$

$$\lim_{x \rightarrow 0^+} f(x)$$

To denominator mein kya $15/40$ dekh kar limit dekheinge

$$n+2=0 \quad | \quad n=-2$$

$\begin{array}{ccccccc} & & & + & + & + & + \\ \hline -\infty & - & - & - & - & - & 2 & & & +\infty \end{array}$

Ex# Find the limits of the following functions.

$$f(x) = \begin{cases} x+1, & x \leq 1 \\ x-1, & x > 1 \end{cases}$$

a) $\lim_{x \rightarrow 1^-} f(x)$

sol L.H.L

$$\lim_{x \rightarrow 1^-} f(x) \Rightarrow 1+1=2$$

b) $\lim_{x \rightarrow 1^+} f(x)$

sol R.H.L

$$\lim_{x \rightarrow 1^+} f(x) = 1-1=0$$

c) $\lim_{x \rightarrow 1} f(x)$

L.H.L $\neq$ R.H.L

Ex# Let $f(x) = \begin{cases} x-1, & x \leq 3 \\ 3x-7, & x > 3 \end{cases}$ Find the limit when $x \rightarrow 3$

L.H.L

$$\lim_{x \rightarrow 3^-} f(x) = x-1 = 3-1=2$$

R.H.L

$$\lim_{x \rightarrow 3^+} f(x) = 3(3)-7=2$$

Ex# Let $g(t) = \begin{cases} t-2, & t < 0 \\ t^2, & 0 \leq t \leq 2 \\ 2t, & t > 2 \end{cases}$

Find (a) $\lim_{t \rightarrow 0} g(t)$

(b) $\lim_{t \rightarrow 1} g(t)$

(c) $\lim_{t \rightarrow 2} g(t)$

$t=0$

L.H.L. $\Rightarrow \lim_{t \rightarrow 0^-} g(t) = 0 - 2 = -2$

R.H.L. $\Rightarrow \lim_{t \rightarrow 0^+} g(t) = 0^2 = 0$

$t=1$

L.H.L. $\Rightarrow \lim_{t \rightarrow 1^-} g(t) = 1^2 = 1$

R.H.L. $\Rightarrow \lim_{t \rightarrow 1^+} g(t) = 1^2 = 1$

$t=2$

L.H.L. $\Rightarrow \lim_{t \rightarrow 2^-} g(t) = 2^2 = 4$

R.H.L. $\Rightarrow \lim_{t \rightarrow 2^+} g(t) = 2(2) = 4$

Ex#2 Show that the function is continuous at $x=0$

$$\text{sol } |x| = \begin{cases} -x, & x < 0 \\ 0, & x = 0 \\ x, & x > 0 \end{cases}$$

L.H.L

$$\lim_{x \rightarrow 0^-} -x \Rightarrow \lim_{x \rightarrow 0^-} |-(0)| = 0$$

R.H.L

$$\lim_{x \rightarrow 0^+} x = \lim_{x \rightarrow 0^+} |0| = 0$$

value of function:-

$$f(0) = 0$$

Hence, the given function is continuous at $x=0$

$$\lim_{x \rightarrow 0} f(x) = 0$$

Ex#3 Discuss the continuity of the function at $x=4$.

$$f(x) = \begin{cases} 2x+3, & x \leq 4 \\ 7+\frac{16}{x}, & x > 4 \end{cases}$$

L.H.L

$$\lim_{x \rightarrow 4^-} 2x+3 = \lim_{x \rightarrow 4^-} 2(4)+3 = 11$$

R.H.L

$$\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \frac{7x+16}{4} = \frac{28+16}{4} = \frac{44}{4} = 11$$

Value of function:-

$$f(4) = 2(4) + 3 = 11$$

Comparison:-

$$f(4) = \lim_{x \rightarrow 4} f(x) = 11$$

Hence, the given function is continuous at $x=4$ Ex#4. Discuss the continuity of the $f(x) = \begin{cases} 3/x-1, & x \neq 1 \\ 3, & x=1 \end{cases}$

sol limit doesn't exist

Ex#5 Find a value of constant k , if any, that will make the function continuous everywhere.

$$f(x) = \begin{cases} 7x-2, & x \leq 1 \\ kx^2, & x > 1 \end{cases}$$

sol L.H.L

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 7x-2 = \lim_{x \rightarrow 1^-} 7(1)-2 = 7-2 = 5$$

R.H.L

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} kx^2 = \lim_{x \rightarrow 1^+} k(1)^2 = k$$

For Continuous Function

$$L.H.L = R.H.L$$

$$5 = k$$

L'HOPITALS' RULE :-

i) Indeterminate forms of type $(0/0)$

Ex#1. Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \left(\frac{0}{0} \right)$

$\therefore$ is derivative lena parega

sol $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \left(\frac{0}{0} \right)$

$$\frac{D_x - 0}{1 - 0} = \frac{2x}{1} = 2(1) = 2$$

Ex#2. Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x} \left(\frac{0}{0} \right)$

sol $\lim_{x \rightarrow 0} \frac{\sin x}{x} \Rightarrow \frac{\cos x}{1} = \cos(0) = 1$

Ex#3 Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$

$$\lim_{x \rightarrow 0} \frac{\cos 2x}{1} = \frac{2 \cos 2x}{2} = 2 \cos 2(0) = 2$$

b) $\lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{\cos x}$

$$\lim_{x \rightarrow \pi/2} \frac{0 - \cos x}{-\sin x} = \frac{-\cos x}{-\sin x} = \frac{0}{1} = 0$$

c) $\lim_{x \rightarrow 0} \frac{e^x - 1}{3x^2}$

$$\lim_{x \rightarrow 0} \frac{e^x}{3x^2} \Rightarrow \frac{e^x - 0}{3x^2} \Rightarrow \frac{e^x}{6x} \Rightarrow \frac{e^0}{6(0)} = \frac{1}{0} = \frac{1}{0}$$

$$d) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$\text{sol } \lim_{x \rightarrow 0} \frac{\sin x}{2x} \Rightarrow \frac{\cos x}{2} = \frac{1}{2}$$

$$e) \lim_{x \rightarrow 0} \frac{\tan x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{\cos x} \div 2x$$

$$\lim_{x \rightarrow 0} \frac{\sec^2 x}{2x} \Rightarrow \cancel{x} \sec^2 x \tan x \Rightarrow \sec^2(0) \tan(0)$$

$$f) \lim_{x \rightarrow \infty} \frac{x^{-4/3}}{\sin(1/x)}$$

$$\text{Ex\#3 Evaluate } \lim_{x \rightarrow 0} \frac{\ln x}{\operatorname{cosec} x} \left(\frac{\infty}{\infty} \right)$$

$$\sin x = 1/\operatorname{cosec} x$$

$$\text{sol } \lim_{x \rightarrow 0} \frac{\ln x}{\operatorname{cosec} x} = \lim_{x \rightarrow 0} \frac{1/x}{-\operatorname{cosec} x \cot x}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin x}{x \cot x}$$

$$= \left(\lim_{x \rightarrow 0} \frac{-\sin x}{x} \right) \cdot \left(\lim_{x \rightarrow 0} \tan x \right)$$

$$= (-1)(0) = 0$$

INDETERMINATE FORMS OF TYPE :- $(0 \cdot \infty \text{ or } \infty \cdot 0)$

$$\text{Ex\#4 Evaluate } \lim_{x \rightarrow 0} x \ln x$$

$$\lim_{x \rightarrow 0} x \ln x = \lim_{x \rightarrow 0} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0} (-x) = 0$$

SEQUENCE:-

A sequence is a function whose domain is a set of integers

EXAMPLES:-

$$1) \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots, \frac{n}{n+1}, \dots, \left[\frac{n}{n+1} \right]_{n=1}^{\infty}$$

$$2) \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots, \frac{1}{2^n}, \dots, \left[\frac{1}{2^n} \right]_{n=1}^{\infty}$$

$$3) \frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \dots, (-1)^{n+1} \frac{n}{n+1}, \dots, \left[(-1)^{n+1} \frac{n}{n+1} \right]_{n=1}^{\infty}$$

$$4) 1, 3, 5, 7, \dots, 2n-1, \dots, \left[2n-1 \right]_{n=1}^{\infty}$$

$$5) 1, \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \dots, \left[\frac{1}{2^{n-1}} \right]_{n=1}^{\infty} \text{ or } \left[\frac{1}{2^n} \right]_{n=0}^{\infty}$$

LIMITS OF SEQUENCE:-

if the n th term u_n of a sequence tends to " l " as n tends to infinity then " l " is called limit of the sequence and we write.

$$\boxed{\lim_{n \rightarrow \infty} u_n = l}$$

Ex#1: Discuss the convergence of divergence of the following sequences.

a) $\{U_n\} = \frac{3n}{n+1}$

sol The n^{th} term of the sequence is.

$U_n = \frac{3n}{n+1}$, then

$\therefore \frac{3n}{n(1+1/n)} \quad \therefore \frac{1}{\infty} = 0$

$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{3n}{n+1} = \lim_{n \rightarrow \infty} \frac{3}{1+\frac{1}{n}} = 3$

Hence, the given sequence is convergent and converges to 3.

b) $\{U_n\} = \frac{n^2}{n^2+9}$

sol The n^{th} term of the sequence is:

$U_n = \frac{n^2}{n^2+9}$, then

$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+9} = \frac{n^2}{n^2(1+\frac{9}{n^2})} = \frac{1}{(1+\frac{9}{n^2})} = 1$

Hence, the given series is convergent and converges to 1.

c) $\{U_n\} = \frac{5n+3}{3n+2}$

sol The n^{th} term of the sequence is.

$U_n = \frac{5n+3}{3n+2}$, then,

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{5n+3}{3n+2} = \lim_{n \rightarrow \infty} \frac{n(5+3/n)}{n(3+2/n)} = \frac{5}{3}$$

Hence, the given series is convergent and converges to $5/3$.

Ex#2 Discuss the divergence or convergence of the following sequence.

a) $\{U_n\} = n^2$

sol $\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} n^2 = \infty$

b) $\{U_n\} = (-1)^n$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} (-1)^n = \begin{cases} 1, & n \text{ is even} \\ -1, & n \text{ is odd} \end{cases}$$

c) $\{U_n\} = (-1)^n \cdot n$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} (-1)^n \cdot n = \begin{cases} \infty, & n \text{ is even} \\ -\infty, & n \text{ is odd} \end{cases}$$

Hence the given series is divergent.

Ex#3

$$a) \{U_n\} = n+1$$

sol

$$b) \{(-1)^{n+1}\} = \begin{cases} 1, & n \text{ is odd} \\ -1, & n \text{ is even} \end{cases}$$

sol

$$c) \{U_n\} = \frac{n}{n+1}$$

sol

$$d) \{U_n\} = 1 + \left(-\frac{1}{2}\right)^n$$

sol

Ex#4

$$a) U_n = \frac{(-1)^n}{n}$$

$$\therefore 1, 3, 5, 7, \dots, 2n-1$$

$$\therefore 2, 4, 6, 8, \dots, 2n$$

sol (The n th odd term is:

$$U_{2n-1} = -1$$

$$\lim_{n \rightarrow \infty} U_{2n-1} = \lim_{n \rightarrow \infty} \frac{-1}{2n-1} = 0$$

The odd term sequence is convergent and converges to 0.

(The n th even term is:

$$U_{2n} = \frac{1}{2n}, \text{ then,}$$

$$\lim_{n \rightarrow \infty} U_{2n} = \lim_{n \rightarrow \infty} \frac{1}{2n} = 0$$

The even term sequence is convergent and converges to 0.

Hence, the given sequence is convergent.

$$\text{Ex \#5. } \{U_n\} = \frac{(-1)^n 2n^3}{n^3+1}$$

sol The n th odd term is

$$U_{2n-1} = \frac{-2(2n-1)^3}{(2n-1)^3+1}, \text{ then,}$$

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$$\lim_{n \rightarrow \infty} U_{2n-1} = \lim_{n \rightarrow \infty} \frac{-2(2n-1)^3}{(2n-1)^3 + 1} = \lim_{n \rightarrow \infty} \frac{-2 \left[\frac{1-1}{(2n)^3} \right]}{\left[\frac{1-2n+1}{(2n)^3} \right]} = -2$$

Continue ...

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Ex#5. Evaluate: $\lim_{x \rightarrow \pi/4} (1 - \tan x) \sec 2x \cdot (0 \cdot \infty)$

$$\begin{aligned} \lim_{x \rightarrow \pi/4} (1 - \tan x) \sec 2x &= \lim_{x \rightarrow \pi/4} \frac{(1 - \tan x)}{\cos 2x} \\ &= \lim_{x \rightarrow \pi/4} \frac{-\sec^2 x}{-2 \sin 2x} = \frac{-\sec^2(\pi/4)}{-2 \sin 2(\pi/4)} = \frac{-2}{-2} = 1 \end{aligned}$$

INDETERMINATE FORMS OF TYPE $(\infty - \infty)$

Ex#6 Evaluate: $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right) (\infty - \infty)$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{\sin x - x}{x \sin x} \right) \left(\frac{0}{0} \right) \Rightarrow \lim_{x \rightarrow 0} \left(\frac{\sin x - x}{x \sin x} \right) \quad \therefore u \cdot v =$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\frac{\sin x - \cos x - 1}{x \cos x + \sin x (1)} \right) \Rightarrow \frac{\cos x - 1}{-x \sin x + \cos x + \sin x}$$

$$\Rightarrow \frac{-\sin x}{-x \sin x + 2 \cos x} \Rightarrow \frac{0}{-0 \cdot \sin(0) + 2(1)} = \frac{0}{2} \Rightarrow 0$$

INDETERMINATE FORMS OF TYPE $0^0, \infty^0, 1^\infty$

Ex#7 $\lim_{x \rightarrow 0} (1 + \sin x)^{1/x}$

$$\text{Sol. } \lim_{x \rightarrow 0} (1 + \sin x)^{1/x} \Rightarrow \lim_{x \rightarrow 0} e^{\left[\ln (1 + \sin x)^{1/x} \right]}$$

$$\Rightarrow e \left[\lim_{x \rightarrow 0} \frac{1}{x} \cdot \ln (1 + \sin x) \right]$$

$$\Rightarrow e \left[\lim_{x \rightarrow 0} \frac{1}{1} \cdot \frac{0 + \cos x}{(1 + \sin x)} \right]$$

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$$\Rightarrow e \left[\lim_{x \rightarrow 0} \left(\frac{\cos x}{1 + \sin x} \right) \right]$$

$$\Rightarrow e \frac{\cos(0)}{1 + \sin(0)}$$

$$\Rightarrow e \left(\frac{1}{1+0} \right)$$

$$\Rightarrow e^1$$

$$\Rightarrow e$$

LEIBNITZ THEOREM:-

$$(f \cdot g)^{(n)} = f^{(n)}g + {}^nC_1 f^{(n-1)}g^{(1)} + {}^nC_2 f^{(n-2)}g^{(2)} + \dots + {}^nC_{n-1} f^{(1)}g^{(n-1)} + f \cdot g^{(n)}$$

Ex#1 Differentiate four times of $x^4 \sin x$ w.r.t. x .

sol let $f(x) = \sin x$, $g(x) = x^4$, then,

$$f^{(1)} = \cos x$$

$$g^{(1)} = 4x^3$$

$$f^{(2)} = -\sin x$$

$$g^{(2)} = 12x^2$$

$$f^{(3)} = -\cos x$$

$$g^{(3)} = 24x$$

$$f^{(4)} = \sin x$$

$$g^{(4)} = 24$$

Using Leibnitz theorem:-

$$(f \cdot g)^{(4)} = f^{(4)}g + {}^4C_1 f^{(4-1)}g^{(1)} + {}^4C_2 f^{(4-2)}g^{(2)} + {}^4C_3 f^{(4-3)}g^{(3)} + {}^4C_4 f^{(4-4)}g^{(4)}$$

$$= (\sin x)(x^4) + 4(-\cos x)(4x^3) + 6(-\sin x)(12x^2) + 4(\cos x)(24x) + 1(\sin x)(24)$$

$$(x^4 \sin x)^{(4)} = x^4 \sin x - 16x^3 \cos x - 72x^2 \sin x + 96x \cos x + 24 \sin x$$

Ex#2. If $y = e^{\tan^{-1}x}$, prove that:

$$(1+x^2)y^{(n+2)} + [2(n+1) \cdot x - 1]y^{(n+1)} + n(n+1)y^{(n)} = 0$$

sol.

$$y = e^{\tan^{-1}x}$$

Diff w.r.t. x

$$y' = e^{\tan^{-1}x} \cdot \frac{1}{1+x^2}$$

$y'(1+x^2) = e^{\tan^{-1}x}$ here $y = e^{\tan^{-1}x}$ and using Product rule.

Again diff w.r.t x

$$2xy' + (1+x^2)y'' = y'$$

$$\therefore f(x) = y''$$

$$(1+x^2)y'' + 2xy' - y' = 0$$

$$(1+x^2)y'' + (2x-1)y' = 0$$

Diff 'n' times, we get,

$$[(1+x^2)y'']^{(n)} + [(2x-1)y']^{(n)} = 0 \quad \text{--- (i)}$$

Using Leibnitz theorem:-

$$f(x)y'' \quad g(x) = (1+x^2)$$

(i) For 1st Term:-

$$\begin{aligned} [y''(1+x^2)]^{(n)} &= y^{(n+2)}(1+x^2) + {}^nC_1 y^{(n+1)}(2x) + {}^nC_2 y^{(n)}(2) \\ &= (1+x^2)y^{(n+2)} + 2nx y^{(n+1)} + n(n-1)y^{(n)} \quad \text{--- (ii)} \end{aligned}$$

ii) For 2nd Term:-

$$\begin{aligned} [(2x-1)y']^{(n)} &= y^{(n+1)}(2x-1) + {}^nC_1 y^{(n)}(2) \\ &= (2x-1)y^{(n+1)} + 2ny^{(n)} \quad \text{--- (iii)} \end{aligned}$$

From eq ii & iii in eq (i), we get

$$(1+x^2)y^{(n+2)} + 2nx y^{(n+1)} + n(n-1)y^{(n)} + (2x-1)y^{(n+1)} + 2ny^{(n)} = 0$$

$$(1+x^2)y^{(n+2)} + 2nx y^{(n+1)} + (2x-1)y^{(n+1)} + n(n-1)y^{(n)} + 2ny^{(n)} = 0$$

$$(1+x^2)y^{(n+2)} + (2nx + 2x - 1)y^{(n+1)} + (n(n-1) + 2n)y^{(n)} = 0$$

$$(1+x^2)y^{(n+2)} + [2x(n+1) - 1]y^{(n+1)} + (n^2 - n + 2n)y^{(n)} = 0$$

$$(1+x^2)y^{(n+2)} + [2x(n+1) - 1]y^{(n+1)} + n(n+1)y^{(n)} = 0 \quad \text{Proved}$$

$$y'(1+x^2) = e^{\tan^{-1}x} \rightarrow \text{here}$$

$$y = e^{\tan^{-1}x}$$

and using Product rule.

Again diff w.r.t x

$$2xy' + (1+x^2)y'' = y'$$

$$\therefore f(x) = y''$$

$$(1+x^2)y'' + 2xy' - y' = 0$$

$$(1+x^2)y'' + (2x-1)y' = 0$$

Diff 'n' times, we get,

$$[(1+x^2)y'']^{(n)} + [(2x-1)y']^{(n)} = 0 \quad \text{--- (i)}$$

Using Leibnitz theorem:-

$$f(x)y'' \quad g(x) = (1+x^2)$$

(i) For 1st Term:-

$$\begin{aligned} [y''(1+x^2)]^{(n)} &= y^{(n+2)}(1+x^2) + {}^nC_1 y^{(n+1)}(2x) + {}^nC_2 y^{(n)}(2) \\ &= (1+x^2)y^{(n+2)} + 2nx y^{(n+1)} + n(n-1)y^{(n)} \quad \text{--- (ii)} \end{aligned}$$

ii) For 2nd Term:-

$$\begin{aligned} [(2x-1)y']^{(n)} &= y^{(n+1)}(2x-1) + {}^nC_1 y^{(n)}(2) \\ &= (2x-1)y^{(n+1)} + 2ny^{(n)} \quad \text{--- (iii)} \end{aligned}$$

From eq ii & iii in eq (i), we get

$$(1+x^2)y^{(n+2)} + 2nx y^{(n+1)} + n(n-1)y^{(n)} + (2x-1)y^{(n+1)} + 2ny^{(n)} = 0$$

$$(1+x^2)y^{(n+2)} + 2nx y^{(n+1)} + (2x-1)y^{(n+1)} + n(n-1)y^{(n)} + 2ny^{(n)} = 0$$

$$(1+x^2)y^{(n+2)} + (2nx + 2x - 1)y^{(n+1)} + (n(n-1) + 2n)y^{(n)} = 0$$

$$(1+x^2)y^{(n+2)} + [2x(n+1) - 1]y^{(n+1)} + (n^2 - n + 2n)y^{(n)} = 0$$

$$(1+x^2)y^{(n+2)} + [2x(n+1) - 1]y^{(n+1)} + n(n+1)y^{(n)} = 0 \quad \text{Proved}$$

TAYLOR'S SERIES:-

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots + \frac{(x-a)^n}{n!} f^{(n)}(a) + \dots$$

MACLAURIN'S SERIES:-Putting $a=0$ in above equation:-

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

Ex#1 Find the Maclaurin Series for:-

a) e^x

Sol The maclaurin Series is given by:

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \frac{x^5}{5!} f^{(5)}(0) + \dots$$

+.... — (i)

we have

$$\begin{aligned} f(x) &= e^x & \Rightarrow f(0) &= 1 \\ f'(x) &= x^1 e^x & f'(0) &= 1 \\ f''(x) &= e^x & f''(0) &= 1 \\ f'''(x) &= e^x & f'''(0) &= 1 \\ f^{(4)}(x) &= e^x & f^{(4)}(0) &= 1 \\ f^{(5)}(x) &= e^x & f^{(5)}(0) &= 1 \end{aligned}$$

Putting values in eq (i)

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \dots$$

b) $\sin x$

The Maclaurin Series is given by:

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \dots \quad (1)$$

we have,

$$f(x) = \sin x \Rightarrow f(0) = 0$$

$$f(x) = \cos x \Rightarrow f'(0) = 1$$

$$f''(x) = -\sin x \Rightarrow f''(0) = 0$$

$$f'''(x) = -\cos x \Rightarrow f'''(0) = -1$$

$$f^{(4)}(x) = \sin x \Rightarrow f^{(4)}(0) = 0$$

$$f^{(5)}(x) = \cos x \Rightarrow f^{(5)}(0) = 1$$

Putting values in eq (1)

 ~~$\sin x$~~

$$\sin x = 0 + x(1) + 0 + \frac{x^3}{6}(-1) + 0 + \frac{x^5}{120}(1)$$

$$\boxed{\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + \dots}$$

c) $\cos x$

The Maclaurin Series is given by:

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \dots \quad (i)$$

we have;

$$f(x) = \cos x \Rightarrow f(0) = 1$$

$$f'(x) = -\sin x \Rightarrow f'(0) = 0$$

RC

$$f''(x) = -\cos x$$

$$\Rightarrow f''(0) = -1$$

$$f'''(x) = \sin x$$

$$\Rightarrow f'''(0) = 0$$

$$f^{(4)}(x) = \cos x$$

$$\Rightarrow f^{(4)}(0) = 1$$

$$f^{(5)}(x) = -\sin x$$

$$\Rightarrow f^{(5)}(0) = 0$$

Put in eq (1)

$$\cos x = 1 + \frac{x}{1!}(0) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}(1) + \dots$$

$$\boxed{\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + \dots}$$

d) $\frac{1}{1-x}$

sol The Maclaurin's Series is given by;

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \dots \quad (1)$$

we have;

$$f(x) = \frac{1}{1-x} \Rightarrow f(0) = 1$$

$$f'(x) = \frac{+1}{(1-x)^2} \Rightarrow f'(0) = +1$$

$$f''(x) = \frac{2}{(1-x)^3} \Rightarrow f''(0) = 2$$

$$f'''(x) = \frac{6}{(1-x)^4} \Rightarrow f'''(0) = 6$$

$$f^{(4)}(x) = \frac{24}{(1-x)^5} \Rightarrow f^{(4)}(0) = 24$$

$$\begin{aligned} &+ 5 \times 24 (1-x)^{-5} \\ &120 (1-x)^{-6} \\ &120 \\ &\overline{(1-x)^6} \end{aligned}$$

$$f^5(x) = \frac{120}{(1-x)^6} \Rightarrow f^6(0) = 120$$

$$(24)(0-1)^{-5}(-5-1)$$

$nx = nx^{n-1}$

put in eq (1)

$$\frac{1}{1-x} = 1 + x + \frac{x^2}{2!}(2) + \frac{x^3}{3!}(6) + \frac{x^4}{4!}(24) + \frac{x^5}{5!}(120) + \dots$$

$$\boxed{\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + x^5 + \dots}$$

Ex#2 Find the Taylor Series for:-

a) $f(x) = e^{-x}$ about $x = \ln 2$

sol

c) $f(x) = \ln x$ about $x=1$

sol The Taylor series is given by:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \frac{f^{(4)}(a)}{4!}(x-a)^4 + \dots \quad (i)$$

we have,

$$f(x) = \ln x$$

$$f'(x) = 1/x$$

$$f''(x) = -1/x^2$$

$$f'''(x) = +2/x^3$$

$$f^{(4)}(x) = -6/x^4$$

$$f^{(5)}(x) = 24/x^5$$

$$\Rightarrow f(1) = 0$$

$$f'(1) = 1$$

$$f''(1) = -1/1 = -1$$

$$f'''(1) = 2$$

$$f^{(4)}(1) = -6$$

$$f^{(5)}(1) = 24$$

$$\therefore \frac{1}{x} = x^{-1} \\ = -1x^{-2} \\ = -1/x^2 \\ = -2x^{-3} \\ = -6x^{-4}$$

Putting values in eq (i)

$$\begin{aligned} \ln x &= 0 + \frac{(x-1)}{1!}(1) + \frac{(x-1)^2}{2}(-1) + \frac{(x-1)^3}{6}(2) + \frac{(x-1)^4}{24}(-6) + \frac{(x-1)^5}{120}(24) + \dots \\ &= (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \frac{(x-1)^5}{5} + \dots \text{Ans.} \end{aligned}$$

d) $f(x) = \sin x$ about $x = \pi/2$

we have,

$$f(x) = \sin x$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

$$f'''(x) = -\cos x$$

$$f^{(4)}(x) = \sin x$$

$$\Rightarrow f(\pi/2) = 1$$

$$\Rightarrow f'(\pi/2) = 0$$

$$\Rightarrow f''(\pi/2) = -1$$

$$\Rightarrow f'''(\pi/2) = 0$$

$$\Rightarrow f^{(4)}(\pi/2) = 1$$

Putting value in Taylor's series -

$$f(x) = f(a) + \frac{(x-a)}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \frac{(x-a)^4}{4!} f^{(4)}(a) + \dots$$

$$\sin x = 1 + 0 + \frac{(x-1)^2}{2!} (-1) + 0 + \frac{(x-1)^4}{4!} (1) + \dots$$

$$\sin x = 1 - \frac{(x-1)^2}{2} + \frac{(x-1)^4}{24} + \dots$$

b) $f(x) = 1/x$ about $x=1$,

sol The Taylor series is given by:

$$f(x) = f(a) + \frac{x-a}{1!} f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \frac{(x-a)^4}{4!} f^{(4)}(a) + \frac{(x-a)^5}{5!} f^{(5)}(a) + \dots \quad (1)$$

we have ! ;

$$f(x) = 1/x$$

$$f'(x) = -1/x^2$$

$$f''(x) = 2/x^3$$

$$f'''(x) = -6/x^4$$

$$f^{(4)}(x) = 24/x^5$$

$$f^{(5)}(x) = -120/x^6$$

$$\Rightarrow f(x) = 1$$

$$\Rightarrow f'(x) = -1$$

$$\Rightarrow f''(x) = 2$$

$$\Rightarrow f'''(x) = -6$$

$$\Rightarrow f^{(4)}(x) = 24$$

$$\Rightarrow f^{(5)}(x) = -120$$

Signatures

put in eq (1)

$$\frac{1}{x} = 1 + \frac{(x-1)}{1!} (-1) + \frac{(x-1)^2}{2!} (2) + \frac{(x-1)^3}{3!} (-6) + \frac{(x-1)^4}{4!} (24) + \frac{(x-1)^5}{5!} (-120) + \dots$$

$$1/x = 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4 - (x-1)^5 + \dots$$

Sequence

A sequence is a function whose domain is a set of integers.

Examples:

1- $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{n}{n+1}, \dots, \left\{ \frac{n}{n+1} \right\}_{n=1}^{\infty}$

2- $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots, \frac{1}{2^n}, \dots, \left\{ \frac{1}{2^n} \right\}_{n=1}^{\infty}$

3- $\frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \dots, (-1)^{n+1} \frac{n}{n+1}, \dots, \left\{ (-1)^{n+1} \frac{n}{n+1} \right\}_{n=1}^{\infty}$

4- $1, 3, 5, 7, \dots, 2n-1, \dots, \{2n-1\}_{n=1}^{\infty}$

5- $1, \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \dots, \left\{ \frac{1}{2^{n+1}} \right\}_{n=1}^{\infty}$ or $\left\{ \frac{1}{2^n} \right\}_{n=0}^{\infty}$

Limit of Sequence.

If the n th term a_n of a sequence tends to " l " as n tends to infinity then " l " is called limit of the sequence and we write

$$\lim_{n \rightarrow \infty} U_n = l$$

Ex#1 Discuss the convergence or divergence of the following sequences.

a), $\{U_n\} = \frac{3n}{n+1}$ b), $\{U_n\} = \frac{n^2}{n^2+9}$ c), $\{U_n\} = \frac{5n+3}{3n+2}$

Sol(a) The n th term of the sequence is

$$U_n = \frac{3n}{n+1}, \text{ then}$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{3n}{n+1} = \lim_{n \rightarrow \infty} \frac{3n \cdot \frac{1}{n}}{n(1 + \frac{1}{n})} = \lim_{n \rightarrow \infty} \frac{3}{1 + \frac{1}{n}} = 3$$

Hence, the given sequence is convergent & converges to 3.

Sol(b) The n th term of sequence is

$$U_n = \frac{n^2}{n^2+9}, \text{ then}$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+9} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{9}{n^2}} = 1$$

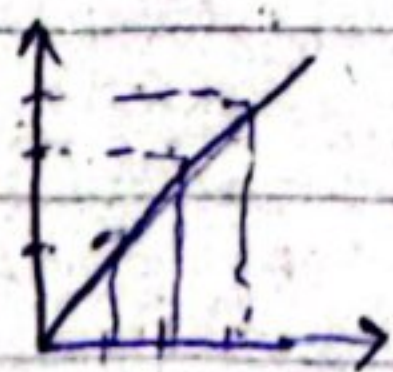
Hence, the given series is convergent & converges to 1.

Sol(c): The n th term of sequence is:

$$U_n = \frac{5n+3}{3n+2}, \text{ then,}$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{5n+3}{3n+2} = \lim_{n \rightarrow \infty} \frac{5 + \frac{3}{n}}{3 + \frac{2}{n}} = \frac{5}{3}$$

Hence, the given series is convergent & converges to $\frac{5}{3}$.



If functions tends to infinity then
Function is divergent.

Ex#2: Discuss -

a) $\{U_n\} = n^2$, b) $\{U_n\} = (-1)^n$, c) $\{U_n\} = (-1)^n \cdot n$

a) $\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} n^2 = \infty$ (Divergent)

b) $\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} (-1)^n = \begin{cases} 1, & n \text{ is even} \\ -1, & n \text{ is odd} \end{cases}$

Hence the given series is divergent (oscillatory)

because we do not reach to a point.

c) $\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} (-1)^n \cdot n = \begin{cases} n, & n \text{ is even} \\ -n, & n \text{ is odd} \end{cases}$

Hence the given series is divergent.

Ex#3 a) $\{U_n\} = n+1$ b) $\{(-1)^{n+1}\}$

c) $\{U_n\} = \frac{n}{n+1}$ d) $\{U_n\} = 1 + \left(-\frac{1}{2}\right)^n$

a) $\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} n+1 = \infty$ (Divergent)

b) $\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} (-1)^{n+1} = \begin{cases} 1, & n \text{ is odd} \\ -1, & n \text{ is even} \end{cases}$

c) $\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = 1$ (Convergent)

d) $\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} 1 + \left(-\frac{1}{2}\right)^n$ For n^{th} even term:

$$\lim_{n \rightarrow \infty} 1 + \left(-\frac{1}{2}\right)^{2n-1} = 1 + 0 = 1$$

For n^{th} odd term.

$$\lim_{n \rightarrow \infty} 1 + \left(-\frac{1}{2}\right)^{2n} = 1 + 0 = 1$$

Hence the given series.

Ex # 4

odd sequence : 1, 3, 5, 7, ..., $2n-1$

Even sequence : 2, 4, 6, 8, ..., $2n$

$$a) U_n = \frac{(-1)^n}{n}$$

Sol:

1) The n th odd term is

$$U_{2n-1} = \frac{-1}{2n-1}, \text{ then,}$$

$$\lim_{n \rightarrow \infty} U_{2n-1} = \lim_{n \rightarrow \infty} \frac{-1}{2n-1} = 0$$

The odd term sequence is convergent and converges to 0.

2) The n th even term is

$$U_{2n} = \frac{1}{2n}, \text{ then,}$$

$$\lim_{n \rightarrow \infty} U_{2n} = \lim_{n \rightarrow \infty} \frac{1}{2n} = 0$$

The even term sequence is convergent & converges to 0.

Hence, the given sequence is convergent because odd and even term is convergent to same value.

$$\text{Ex \# 5, } \{U_n\} = \frac{(-1)^n 2n^3}{n^3+1}$$

1) The n th odd term is

$$U_{2n-1} = \frac{-2(2n-1)^3}{(2n-1)^3+1}, \text{ then,}$$

$$\lim_{n \rightarrow \infty} U_{2n-1} = \lim_{n \rightarrow \infty} \frac{-2(2n-1)^3}{(2n-1)^3+1} = \lim_{n \rightarrow \infty} \frac{-2 \left[1 - \frac{1}{(2n)^3} \right] 2n^3}{\left[1 - \frac{1}{(2n)^3} + \frac{1}{(2n)^3} \right] 2n^3} = -2$$

4) The n th even term is

$$U_{2n} = \frac{2(2n)^3}{(2n)^3 + 1} = \frac{16n^3}{8n^3 + 1}$$

$$\lim_{n \rightarrow \infty} U_{2n} = \lim_{n \rightarrow \infty} \frac{16n^3}{8n^3 + 1} = \lim_{n \rightarrow \infty} \frac{16x^3}{x^3(8 + 1/n^3)} = \lim_{n \rightarrow \infty} \frac{16}{8 + 1/n^3}$$

$$\lim_{n \rightarrow \infty} U_{2n} = 2.$$

Leibnitz Theorem

$$(fg)^{(n)} = f^{(n)}g + {}^nC_1 f^{(n-1)}g^{(1)} + {}^nC_2 f^{(n-2)}g^{(2)} + \dots + {}^nC_{n-1} f^{(1)}g^{(n-1)} + fg^{(n)}$$

Note (g is the term which differentiates repeatedly to 0)

Ex #1 Differentiate four times of $x^4 \sin x$ w.r.t x .

Sol:

Let $f(x) = \sin x$ & $g(x) = x^4$, then

$$f^{(1)} = \cos x$$

$$g^{(1)} = 4x^3$$

$$f^{(2)} = -\sin x$$

$$g^{(2)} = 12x^2$$

$$f^{(3)} = -\cos x$$

$$g^{(3)} = 24x$$

$$f^{(4)} = \sin x$$

$$g^{(4)} = 24$$

Using Leibnitz theorem.

$$\begin{aligned} (f \cdot g)^{(4)} &= f^{(4)}g + {}^4C_1 f^{(3)}g^{(1)} + {}^4C_2 f^{(2)}g^{(2)} + {}^4C_3 f^{(1)}g^{(3)} + fg^{(4)} \\ &= (\sin x)(x^4) + 4(-\cos x)(4x^3) + 6(-\sin x)(12x^2) + 4\cos x(24x) \\ &\quad + \sin x(24) \end{aligned}$$

$$(x^4 \sin x)^{(4)} = x^4 \sin x - 16x^3 \cos x - 72x^2 \sin x + 96x \cos x + 24 \sin x$$

Ex #2 If $y = e^{ax^2}$, prove that

$$(1+x^2)y^{(n+2)} + [2(n+1)x-1]y^{(n+1)} + n(n+1)y^{(n)} = 0$$